# Electrostatic ballooning of spiders with multiple charged silk threads in near-surface turbulence: analytical limits and take-off probability

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We start from the three-dimensional model of Habchi and Jawed [1], in which electrically charged silk threads are treated as discrete elastic rods subject to the atmospheric electric field, Coulomb repulsion, and air drag in the resistive force theory (RFT) approximation. We complete the model with the atmospheric field profile and experimentally measured charges [2, 3], and we place the bundle in synthetic near-surface turbulence generated by kinematic simulation with surface-layer statistics. Take-off becomes a stochastic event, characterized by a take-off probability; in the limit of eddies much larger and slower than the bundle, it reduces to a drift–diffusion result that smooths the deterministic threshold of the normalized lift-to-weight ratio, $\bar{F}_l = 1$, into a sigmoid controlled by a Péclet number. The simulator is verified against analytical terminal velocities and against the still-air steady state of tip-charged threads, for which the equilibrium equations have an exact first integral and, in the small-angle limit, a closed-form opening law; the latter is also the still-air reference for the opening of the bundle in turbulence. Ensembles of 8200 simulations for $N \leq 8$ show that the take-off probability collapses on a single function of the scaled drift for all $N$ and turbulence intensities: the bundle enters only through its settling speed, and the drift–diffusion form holds once the eddy diffusivity is replaced by the Lagrangian diffusivity of the flow. Finally, we train a graph network simulator on the simulation data and test its generalization to numbers of threads and turbulence intensities not seen during training. It is more accurate than the reference methods over the first second after release, with a thread-position error of 19 mm against 157 mm for passive advection by the air, but its rollouts become unphysical within about 1 s. Its take-off probabilities are therefore less accurate than those of a constant-velocity extrapolation ($|\Delta P| = 0.45$ against 0.14), even though it runs 47–225 times faster than the simulator per simulated second.

## I. INTRODUCTION

Passive aerial dispersal of spiders (*ballooning*) is explained by two competing mechanisms: aerodynamic drag in rising air currents, and the electrostatic force exerted by the atmospheric potential gradient on negatively charged silk threads [2, 4]. From experiments in a chamber with a controlled electric field, Morley and Gorham estimated a total thread charge of about 1.15 nC for spiders of 0.9 mg [3]. Habchi and Jawed built the first three-dimensional model with multiple threads and showed that the Coulomb repulsion between threads produces a conical opening of the bundle [1]. Their model, however, assumes still air, although ballooning starts in the turbulent surface layer of the atmosphere.

The aerodynamic view has a long history. Humphrey [21] used a simple mechanical model of the spider–filament system to delimit, in terms of spider mass and silk length, the region in which ballooning is possible, and wind-tunnel experiments by Suter [22] showed that both the silk and the body contribute to the drag. Reynolds *et al.* [23] modelled the silk as a fully flexible, extensible line in turbulent flow and found that turbulence contorts the silk, so that ballooners have little control over their drag once airborne. More recently, Cho *et al.* observed large crab spiders (16–20 mg) taking off with 50–60 nanoscale fibres in weak updrafts of 0.1–0.5 m/s [9], and a bead–spring model with anisotropic drag was used to study the passive dynamics of a ballooning filament in shear, vortical, and turbulent flows [24]. None of these models includes electric charge. The electrostatic view received strong support when Morley and Robert showed that *Erigone* spiders detect vertical electric fields of atmospheric strength, that such fields alone elicit pre-ballooning behaviour and take-off in still air, and that the sensory hairs (trichobothria) respond mechanically to them [4]. Field-chamber experiments on three linyphiid species, however, found that wind had a much stronger

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effect on take-off than the electric field, the combination of the two giving the highest dispersal rates [11]; a recent review concludes that the relative importance of the two mechanisms is still open [12].

On the modelling side, thin elastic filaments are commonly simulated with the discrete elastic rods (DER) method [7], which represents a rod by its centreline nodes and edge twist angles and derives the forces from discrete stretching, bending, and twisting energies. Implicit time integration of DER makes very slender, nearly inextensible filaments tractable [1], and open-source implementations are available [8]. Habchi and Jawed coupled DER with resistive force theory and pairwise Coulomb repulsion, which is the starting point of the present work [1]. The cost of such simulations grows quickly with the number of threads, because the Coulomb interaction couples every pair of charged nodes and makes the Jacobian dense.

On the computational side, simulating many interacting charged filaments is expensive because the Coulomb interaction couples all nodes. Learned simulators based on graph neural networks reproduce the dynamics of particles at a fraction of the cost [20], and have been used to accelerate implicit solvers for rod dynamics [25], and long-range message-passing schemes such as Ewald message passing were designed precisely for $1/r$ interactions [15]. Whether such models generalize to filament bundles with more threads than seen during training has not been tested. Mesh-based graph networks extend the approach to cloth and deformable solids [16].

**The contributions of this work** are:

1. the multi-thread charged model of Habchi and Jawed placed in realistic synthetic near-surface turbulence, with take-off defined as a stochastic event;

2. an analytical drift–diffusion reference for the take-off probability and its comparison with ensemble simulations, which isolates the effect of the finite size and deformation of the bundle;

3. a learned graph network simulator trained on the ensembles, with a test of its generalization to unseen numbers of threads and turbulence intensities, and a comparison of its cost with that of the simulator.

The paper is organized as follows. Section II presents the model. Section III describes the air flow and the take-off probability, and Sec. IV gives the still-air analytical solution for tip-charged threads, used to verify the simulator. Section V describes the numerical method and Sec. VI the learned surrogate. Results are presented in Sec. VII and discussed in Sec. VIII; Sec. IX concludes.

## II. PHYSICAL MODEL

### A. Geometry and parameters

The spider is a sphere of radius $r_s$, mass $m$, and charge $Q_s$, to which $N$ identical threads of length $L$, radius $r$, and density $\rho_t$ are attached. Each thread carries a charge $q$ (in absolute value). From the motion of ballooning spiders in a controlled field, Morley and Gorham infer that the charge is not uniformly distributed and resides mainly on the upper portion of the silk [3]; the distribution itself was not measured. We idealize it as a charge concentrated at the thread tip, one of the two limiting cases of [1].

We use the following sign convention. The thread charge is negative and the fair-weather atmospheric field points downward, so the electrostatic force on the thread is *upward*. We work with the positive quantities $q$ and $E(z)$; the force on the charge $q$ is $+qE(z)\,\hat{\boldsymbol{z}}$.

### B. Electric field

Over flat ground the fair-weather field is $E_0 \approx 100$–$120$ V/m [2] and is practically constant on the scale of one thread.

For the experiment of Morley and Gorham, with launch from an aluminum-foil-covered prominence, we use the fit given in [1]:

$$E(z) = E_1 e^{-z/z_1} + E_2 e^{-z/z_2} + E_\infty, \tag{1}$$

with $E_1 = 2.52 \times 10^5$ V/m, $z_1 = 1.51$ mm, $E_2 = 5.07 \times 10^4$ V/m, $z_2 = 7.93$ mm, and $E_\infty = 7.41 \times 10^3$ V/m. The two exponential terms decay within a few millimetres of the prominence. We use the full profile of Eq. (1) only to reproduce the chamber experiment (Test 1, Sec. VII A); all other simulations, including the turbulence ensembles, use the height-independent field $E = E_\infty$.

TABLE I. Reference parameters. Spider and thread values are those of [1].

| Quantity | Symbol | Value | Source |
|---|---|---|---|
| Spider mass | $m$ | 1 mg (0.1–10 mg in Table II) | [1] |
| Spider radius | $r_s$ | 1 mm | [1] |
| Spider charge | $Q_s$ | 3 pC (negligible) | [1, 3] |
| Thread radius | $r$ | 300 nm | [1] |
| Thread length | $L$ | 0.5 m | [1, 3] |
| Silk density | $\rho_t$ | 1200 kg/m$^3$ | [1] |
| Young's modulus | $Y$ | 5–50 GPa (typical 25) | [1] |
| Air viscosity | $\mu$ | $1.837 \times 10^{-5}$ Pa s | |
| Field over flat ground | $E_0$ | 100–120 V/m | [2] |
| Field near a prominence | $E(z)$ | Eq. (1) (Test 1); $E_\infty = 7.41 \times 10^3$ V/m elsewhere | [1, 3] |
| Charge per thread | $q$ | 0.1–10 nC | [3] |

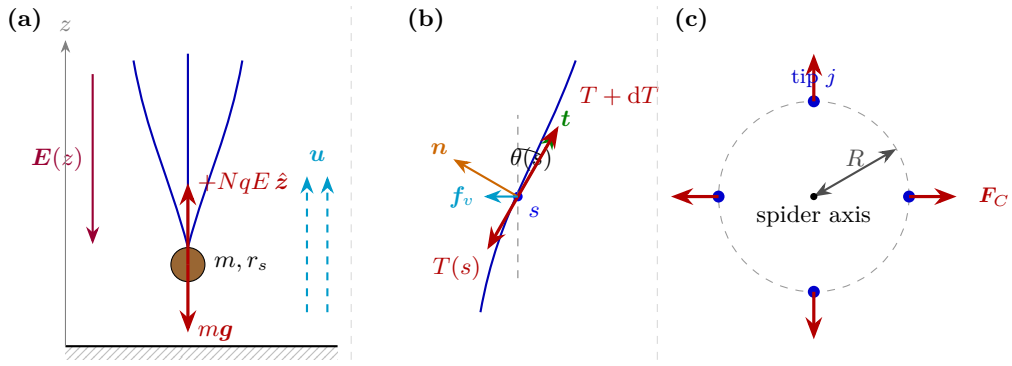



FIG. 1. (a) Schematic representation of the spider–silk system in atmospheric turbulence and a vertical electric field $\boldsymbol{E}(z)$. (b) Geometry and force equilibrium along a single flexible thread element at arc length $s$, showing local tangent $\boldsymbol{t}$, normal $\boldsymbol{n}$, angle $\theta(s)$, tension $T(s)$, and RFT drag $\boldsymbol{f}_v$. (c) Top view at $s = L$ for $N = 4$ tip-charged threads, showing the conical opening of radius $R$ driven by mutual radial Coulomb repulsion $\boldsymbol{F}_C$.

### C. Forces

Let $s \in [0, L]$ be the arc length along a thread ($s = 0$ at the spider), $\boldsymbol{x}(s, t)$ its position, $\boldsymbol{v} = \partial_t \boldsymbol{x}$ the thread velocity, and $\boldsymbol{t} = \partial_s \boldsymbol{x}$ the local unit tangent vector.

*a. Air drag (RFT).* For a very slender thread at low Reynolds number ($\mathrm{Re} \ll 1$), local air resistance is governed by Resistive Force Theory (RFT) [1]. The drag force per unit length depends on the relative velocity $\boldsymbol{v}_{\mathrm{rel}} = \boldsymbol{v} - \boldsymbol{u}$ between the thread and the fluid, decomposed into components parallel and perpendicular to the local tangent $\boldsymbol{t}$:

$$\boldsymbol{f}_v = -\eta_{\parallel}(\boldsymbol{v}_{\mathrm{rel}} \cdot \boldsymbol{t})\boldsymbol{t} - \eta_{\perp}\left[\boldsymbol{v}_{\mathrm{rel}} - (\boldsymbol{v}_{\mathrm{rel}} \cdot \boldsymbol{t})\boldsymbol{t}\right] = -\left[\eta_{\parallel}\, \boldsymbol{t}\boldsymbol{t}^{T} + \eta_{\perp}\left(\mathbb{I} - \boldsymbol{t}\boldsymbol{t}^{T}\right)\right] \boldsymbol{v}_{\mathrm{rel}}. \tag{2}$$

Here, $\boldsymbol{t}\boldsymbol{t}^{T} = \boldsymbol{t} \otimes \boldsymbol{t}$ denotes the outer product projecting onto the local tangent direction.

$$\eta_{\parallel} = \frac{2\pi\mu}{\ln(L/r) - \frac{1}{2}}, \qquad \eta_{\perp} = \frac{4\pi\mu}{\ln(L/r) + \frac{1}{2}}. \tag{3}$$

Here $\boldsymbol{v} = \partial_t \boldsymbol{x}$ is the thread velocity, $\boldsymbol{u}$ is the local air velocity. **The velocity $\boldsymbol{u}$ is our extension**: in [1], $\boldsymbol{u} = 0$. For the parameters of Table I, $\ln(L/r) \approx 14.3$, $\eta_{\parallel} \approx 8.35 \times 10^{-6}$ Pa s, and $\eta_{\perp} \approx 1.56 \times 10^{-5}$ Pa s. We define the drag anisotropy ratio:

$$\beta \equiv \frac{\eta_{\perp}}{\eta_{\parallel}} = \frac{2\left[\ln(L/r) - \frac{1}{2}\right]}{\ln(L/r) + \frac{1}{2}} \approx 1.87. \tag{4}$$

The anisotropy $\beta > 1$ directly governs the coupling between thread inclination and tension along the bundle.

The spider body experiences Stokes drag, $\boldsymbol{F}_{v,s} = -6\pi\mu r_s(\boldsymbol{v}_s - \boldsymbol{u})$. We write $\zeta_s \equiv 6\pi\mu r_s \approx 3.5 \times 10^{-7}$ N s/m.

*b. Electrostatic force.* The force $qE(z)\hat{\boldsymbol{z}}$ acts at the tip, $s = L$.

*c. Coulomb repulsion.* The repulsion acts only between the tips ($s = L$):

$$\boldsymbol{F}_{r,\mathrm{tip},k} = k_e q^2 \sum_{i\neq k} \frac{\boldsymbol{x}_k(L) - \boldsymbol{x}_i(L)}{\|\boldsymbol{x}_k(L) - \boldsymbol{x}_i(L)\|^3}, \qquad k_e = \frac{1}{4\pi\varepsilon_0}. \tag{5}$$

*d. Weight.* The thread weight is $N\rho_t \pi r^2 L g \approx 1.7 \times 10^{-9}$ N for $N = 1$, about $10^{-4}$ of the spider weight. We neglect it in the analysis but keep it in the simulations.

### D. Equilibrium equations for a flexible thread

Habchi and Jawed show that bending stiffness is negligible, with dimensionless ratio $\overline{YI} \sim 10^{-10}$ [1]. Each thread is a curved filament whose shape $\boldsymbol{x}(s)$ is part of the solution; straight segments appear only in the discretization of Sec. V. We therefore treat each thread as a perfectly flexible, inextensible string with tension $T(s)$, and neglect the (tiny) thread inertia. Local equilibrium reads

$$\partial_s (T\,\boldsymbol{t}) + \boldsymbol{f}_v = 0, \qquad 0 \le s < L, \tag{6}$$

since, for tip charge, the electrostatic and Coulomb forces act only at $s = L$, through the boundary condition below. The simulations of Sec. V do not use Eq. (6): they solve the full dynamics of the discretized threads. Equation (6) is the starting point of the analytical solution of Sec. IV, against which the simulations are checked.

The boundary conditions are:

- at the spider ($s = 0$): $\displaystyle\sum_{j=1}^{N} T_j(0)\,\boldsymbol{t}_j(0) + (Q_s E - mg)\,\hat{\boldsymbol{z}} + \boldsymbol{F}_{v,s} = m\,\dot{\boldsymbol{v}}_s$;
- at the tip ($s = L$): $T(L)\,\boldsymbol{t}(L) = qE\,\hat{\boldsymbol{z}} + \boldsymbol{F}_{r,\mathrm{tip}}$.

### E. Dimensionless groups

For a nearly vertical bundle ascending at velocity $U$ relative to the air, the relative flow is aligned with the local thread tangents ($\boldsymbol{t} \approx \hat{\boldsymbol{z}}$), making the drag governed primarily by the parallel friction coefficient $\eta_\parallel$. The global vertical force balance reads:

$$NqE - mg = \left(N\eta_\parallel L + \zeta_s\right) U, \tag{7}$$

where $U$ is the characteristic terminal velocity. It defines the two groups used below; $\Lambda$ enters only the still-air opening of Sec. IV:

$$\bar{F_l} = \frac{NqE}{mg} \qquad \text{normalized lift-to-weight ratio (as in [1])}, \tag{8}$$

$$\Lambda = \frac{k_e\, q}{EL^2} = \frac{k_e q^2/L^2}{qE} \qquad \text{ratio of Coulomb repulsion to electrostatic lift per thread.} \tag{9}$$

For typical experimental conditions ($q = 0.6$ nC, $E = 7.41 \times 10^3$ V/m, $L = 0.5$ m), the relative Coulomb interaction parameter evaluates to $\Lambda \approx 2.9 \times 10^{-3}$.

## III. AIR FLOW: MEAN-FLOW REFERENCE AND NEAR-SURFACE TURBULENCE

### A. Uniform flow: a symmetry, not a new regime

In Eq. (2) the air velocity enters only through $\boldsymbol{v} - \boldsymbol{u}$. For a uniform and steady flow, $\boldsymbol{u} = \text{const.}$, the change of frame $\boldsymbol{x}' = \boldsymbol{x} - \boldsymbol{u}\,t$ maps the problem onto still air: accelerations are unchanged, and $E$ varies negligibly over the distance travelled. The mapping is exact at all times if the initial state is the same relative to the air ($\boldsymbol{v}'(0) = \boldsymbol{v}(0) - \boldsymbol{u}$); otherwise the two motions differ during a transient and converge to the same steady state. The shape of the bundle and the velocity $U$ relative to the air are therefore exactly those in still air, and the ground-frame velocity is $V = U + w$ for $\boldsymbol{u} = w\hat{\boldsymbol{z}}$. This symmetry is used only as a test of the code (Fig. 9). A flow can change the shape of the bundle only if it varies in space or in time on scales comparable to $L$ or to the response time of the spider.

### B. Mean-flow reference: deterministic threshold

For a steady updraft $w$, the spider rises if its ground velocity $V = U + w$ is positive. From Eq. (7),

$$V = w_s\,(\bar{F}_l - 1) + w > 0 \quad \Longleftrightarrow \quad w > w_c \equiv w_s\,(1 - \bar{F}_l), \qquad w_s \equiv \frac{mg}{N\eta_\parallel L + \zeta_s}, \tag{10}$$

where $w_s$ is the terminal falling speed of the uncharged spider–bundle system in calm air (obtained by balancing weight $mg$ against total air drag $(N\eta_\parallel L + \zeta_s)w_s$). This straight line is the baseline reference against which the turbulent results of Sec. III C are compared. It assumes nearly vertical threads; Coulomb tilting of the threads (large $\Lambda$, large $N$) increases the total drag and shifts the true take-off boundary.

TABLE II. Mean-flow threshold $w_c$ (m/s) from Eq. (10). $L = 0.5$ m, $r = 300$ nm, $E = 7.41 \times 10^3$ V/m, tip charge. Negative values: the spider rises even in a downdraft. Values above ∼1 m/s exceed the range of validity of the low-Reynolds-number drag laws and should be flagged. The column $q = 0$ equals the settling speed $w_s$. Numerical values: $-V$ at steady state in still air ($Q_s = 0$, $N_t = 100$). The charged multi-thread cases lie up to 12% below the analytical value (in absolute value) because the Coulomb splay of the threads increases the drag.

| | | $q = 0$ | | $q = 0.6$ nC per thread | |
|---|---|---|---|---|---|
| $N$ | $m$ (mg) | analytical | numerical | analytical | numerical |
| 1 | 0.1 | 0.217 | 0.218 | −0.767 | −0.770 |
| 1 | 1 | 2.17 | 2.18 | 1.19 | 1.19 |
| 1 | 10 | 21.7 | 21.8 | 20.7 | 20.8 |
| 2 | 0.1 | 0.113 | 0.114 | −0.910 | −0.882 |
| 2 | 1 | 1.13 | 1.13 | 0.106 | 0.106 |
| 2 | 10 | 11.3 | 11.3 | 10.3 | 10.3 |
| 4 | 0.1 | 0.058 | 0.058 | −0.986 | −0.909 |
| 4 | 1 | 0.576 | 0.579 | −0.468 | −0.458 |
| 4 | 10 | 5.76 | 5.78 | 4.71 | 4.68 |
| 8 | 0.1 | 0.029 | 0.030 | −1.03 | −0.898 |
| 8 | 1 | 0.291 | 0.293 | −0.763 | −0.708 |
| 8 | 10 | 2.91 | 2.92 | 1.85 | 1.82 |

### C. Near-surface turbulence

*a. Why turbulence, and at which scale.* Ballooning starts within a few metres of the ground, in the atmospheric surface layer. There, the standard deviation of the vertical velocity scales with the friction velocity, $\sigma_w \approx 1.25\,u_*$ in neutral conditions [26, 27], and the dominant eddies have a size of the order of the height above ground, $\ell_w \sim z$ [26]. At take-off heights of 0.5–2 m this is comparable to the thread length $L = 0.5$ m. A realistic flow must therefore vary in space across the bundle, not only in time; a spatially uniform gust $w(t)$ would be artificial.

*b. Synthetic turbulence.* We generate the air velocity by kinematic simulation [28, 29], a sum of $N_k$ random Fourier modes,

$$\boldsymbol{u}(\boldsymbol{x}, t) = U_h\hat{\boldsymbol{x}} + \sum_{n=1}^{N_k} \left[\boldsymbol{a}_n \cos(\boldsymbol{k}_n \cdot \boldsymbol{x}' + \omega_n t) + \boldsymbol{b}_n \sin(\boldsymbol{k}_n \cdot \boldsymbol{x}' + \omega_n t)\right], \qquad \boldsymbol{x}' = \boldsymbol{x} - U_h t\,\hat{\boldsymbol{x}}, \tag{11}$$

with random directions $\hat{\boldsymbol{k}}_n$, amplitudes $\boldsymbol{a}_n, \boldsymbol{b}_n \perp \boldsymbol{k}_n$ (so that $\nabla \cdot \boldsymbol{u} = 0$), and magnitudes set by the energy spectrum in the shell $\Delta k_n$ around $k_n$, normalized so that $\frac{1}{2}\langle|\boldsymbol{u} - U_h\hat{\boldsymbol{x}}|^2\rangle = \int_0^\infty E(k)\,\mathrm{d}k$ [29]. The energy spectrum is of von Kármán form,

$$E(k) \propto \sigma^2 \ell\, \frac{(k\ell)^4}{[1 + (k\ell)^2]^{17/6}}, \qquad \int_0^\infty E(k)\,\mathrm{d}k = \tfrac{3}{2}\sigma^2, \tag{12}$$

which has the $k^{-5/3}$ inertial range. The mean horizontal wind $U_h$ advects the eddies past the spider (Taylor's hypothesis); by the symmetry above it has no other effect. The unsteadiness frequencies are $\omega_n = \lambda\sqrt{k_n^3 E(k_n)}$ with

$\lambda \approx 0.5$ [29]. Each random seed defines one realization; the modes are stored with the trajectory, so that each run can be reproduced exactly.

TABLE III. Turbulence parameters. Surface-layer scalings from [26, 27].

| Quantity | Symbol | Range | Note |
|---|---|---|---|
| Friction velocity | $u_*$ | 0.1–0.3 m/s | light to moderate wind |
| Vertical velocity std. dev. | $\sigma_w$ | $1.25\,u_* \approx 0.12$–$0.38$ m/s | neutral surface layer |
| Integral length scale | $\ell$ | 0.5–2 m | $\sim$ height above ground |
| Mean horizontal wind | $U_h$ | 0.5–3 m/s | 1 m/s in the simulations |
| Number of modes | $N_k$ | 200 | $k_{\min} = 0.1/\ell$, $k_{\max} = 2\pi/(5\,\Delta l)$ |
| Unsteadiness parameter | $\lambda$ | 0.5 | [29] |

*c. Validity.* Relative velocities are of order $\sigma_w \lesssim 0.4$ m/s, so the thread Reynolds number is $\mathrm{Re} = \rho_{\mathrm{air}}\sigma_w r/\mu \sim 10^{-2}$ and RFT remains valid. The model (11) is homogeneous and isotropic, whereas surface-layer turbulence is anisotropic ($\sigma_u > \sigma_w$) and its scale grows with height. We match the vertical statistics, $\sigma = \sigma_w$ and $\ell = \ell_w$, which control take-off.

Figure 6 (50 realizations, $\sigma_w = 0.25$ m/s, $\ell = 1$ m, $N_k = 200$): the sample standard deviation of $w$ is 0.249 m/s after the renormalization of Appendix E, the PDF is Gaussian, and the one-dimensional spectrum follows the von Kármán target, including the $k^{-5/3}$ range, up to $k \approx 80$ m$^{-1}$; above, it falls off towards $k_{\max} = 251$ m$^{-1}$ because of the finite shell width.

### D. Take-off probability

*a. Definition.* The spider is released at height $z_0$ above the ground (tip of a grass blade or twig) and followed until it either reaches a height $h$ (*take-off*) or returns to the ground, $z = 0$ (*failure*). The take-off probability $P$ is the fraction of realizations that reach $h$. It depends on $\bar{F}_l$, $N$, $\Lambda$, and the turbulence through $\sigma_w/w_s$ and $\ell/L$.

*b. Analytical reference: drift–diffusion limit.* If the turbulent eddies are much larger than the bundle ($\ell \gg L$) and their turnover time $\ell/\sigma_w$ is much longer than the velocity adjustment time of the spider (the time constant required to reach terminal speed),

$$t_s = \frac{m}{N\eta_\parallel L + \zeta_s} = \frac{w_s}{g} \approx 0.2 \text{ s} \quad (m = 1 \text{ mg},\ N = 1), \tag{13}$$

the spider simply follows the local fluid motion with a constant downward/upward drift speed $U_0 = w_s(\bar{F}_l - 1)$ relative to the air. On time scales longer than the eddy lifetime $T_L \sim \ell/\sigma_w$, its height $z(t)$ executes a drift–diffusion process with turbulent diffusivity $K = \sigma_w^2 T_L \sim \sigma_w \ell$ [30]. The probability $P(z_0)$ to reach take-off height $h$ before falling back to the ground ($z = 0$) satisfies the steady advection–diffusion equation $U_0 P' + K P'' = 0$ with boundary conditions $P(0) = 0$ and $P(h) = 1$:

$$\boxed{P = \frac{1 - e^{-U_0 z_0/K}}{1 - e^{-U_0 h/K}}, \qquad U_0 = w_s(\bar{F}_l - 1), \quad K \simeq \sigma_w \ell} \tag{14}$$

For $U_0 \to 0$, $P \to z_0/h$; for $|U_0|h/K \gg 1$, $P$ switches sharply from 0 to 1 at $\bar{F}_l = 1$, recovering the deterministic threshold. Turbulence thus smooths the step at $\bar{F}_l = 1$ into a sigmoid of width set by the Péclet number $\mathrm{Pe} = w_s h/K$.

With $w_s = mg/(N\eta_\parallel L + \zeta_s)$, $t_s = w_s/g$, $K = \sigma_w \ell$, and $\mathrm{Pe} = w_s h/K$ ($h = 2$ m), for $\sigma_w \ell$ between $0.12 \times 0.5 = 0.06$ and $0.38 \times 2 = 0.76$ m$^2$/s:

| $m$ (mg) | $N$ | $w_s$ (m/s) | $t_s$ (s) | Pe ($K = 0.06$) | Pe ($K = 0.76$) |
|---|---|---|---|---|---|
| 0.1 | 1 | 0.217 | 0.022 | 7.2 | 0.57 |
| 0.1 | 8 | 0.029 | 0.003 | 0.97 | 0.08 |
| 1 | 1 | 2.17 | 0.22 | 72 | 5.7 |
| 1 | 8 | 0.291 | 0.030 | 9.7 | 0.77 |
| 10 | 1 | 21.7 | 2.2 | 720 | 57 |
| 10 | 8 | 2.91 | 0.30 | 97 | 7.7 |

Turbulence matters (Pe of order 1–10) for light spiders and many threads; for $m = 10$ mg and $N = 1$ the transition is practically deterministic, and $w_s$ exceeds the validity range of the drag laws (Appendix A). The simulations give an effective diffusivity about four times smaller than $\sigma_w \ell$ (Sec. VII C), so the effective Pe is about four times larger than in this table.

## IV. STILL-AIR ANALYTICAL SOLUTION FOR TIP-CHARGED THREADS

The steady state of tip-charged threads in still air has an exact first integral and, for small angles, a closed-form opening. We use both only as references: to verify the simulator (Sec. VII B) and as the still-air value against which the opening of the bundle in turbulence is measured (Sec. VII C).

### A. Equations along one thread

Consider the steady state in still air. In a 2D vertical plane, the thread profile $\boldsymbol{x}(s)$ is parameterized by the local inclination angle $\theta(s)$ relative to the vertical $\hat{\boldsymbol{z}}$, with unit tangent $\boldsymbol{t} = (\sin\theta, \cos\theta)$ and normal $\boldsymbol{n} = (\cos\theta, -\sin\theta)$ such that $\partial_s \boldsymbol{t} = \theta' \boldsymbol{n}$. In the frame of the ascending spider, the relative air velocity flowing past the thread is $\boldsymbol{v} - \boldsymbol{u} = U\hat{\boldsymbol{z}}$.

Decomposing the unit vertical vector as $\hat{\boldsymbol{z}} = (\cos\theta)\boldsymbol{t} - (\sin\theta)\boldsymbol{n}$, the RFT drag force per unit length (2) simplifies to $\boldsymbol{f}_v = -\eta_\parallel U \cos\theta\, \boldsymbol{t} + \eta_\perp U \sin\theta\, \boldsymbol{n}$. Substituting this into the string equilibrium equation (6) and projecting along $\boldsymbol{t}$ and $\boldsymbol{n}$ yields

$$\frac{\mathrm{d}T}{\mathrm{d}s} = \eta_\parallel U \cos\theta, \tag{15}$$

$$T\frac{\mathrm{d}\theta}{\mathrm{d}s} = -\eta_\perp U \sin\theta. \tag{16}$$

### B. An exact first integral

To isolate the spatial variation of thread geometry, we eliminate the arc-length coordinate $s$ by dividing the normal equilibrium equation (16) by the tangential equation (15). Using $\beta = \eta_\perp/\eta_\parallel$, this yields a separable differential equation relating local tension and angle:

$$\frac{\cos\theta}{\sin\theta}\,\mathrm{d}\theta = -\beta\,\frac{\mathrm{d}T}{T}. \tag{17}$$

Integrating both sides directly yields an exact first integral for flexible threads in RFT:

$$\boxed{T(s)^\beta \,\sin\theta(s) = \text{const.}} \tag{18}$$

Equation (18) is exact within RFT, requiring no small-angle approximation.

Figure 4: for $N = 2$, $\bar{F}_l = 2$, $N_t = 100$, $T^\beta \sin\theta$ along the thread varies by 0.019% of its mean (Table VII). Derivation check: from Eq. (16)/Eq. (15), $\cot\theta\,\mathrm{d}\theta = -\beta\,\mathrm{d}T/T$, so $\ln\sin\theta = -\beta\ln T + \text{const.}$

### C. Boundary conditions for $N$ threads

Assume a symmetric configuration: the tips of the $N$ threads lie on a horizontal circle of radius $R$, at azimuths $2\pi j/N$. The radial Coulomb force on one tip due to the other $N-1$ tips is

$$F_C = \frac{k_e q^2}{4R^2}\, S_N, \qquad S_N \equiv \sum_{j=1}^{N-1} \frac{1}{\sin(\pi j/N)}, \tag{19}$$

with $S_2 = 1$, $S_4 = 1 + 2\sqrt{2} \approx 3.83$, and $S_8 \approx 11.2$. It follows from the separation $2R\sin(\pi j/N)$ between the tip considered and tip $j$; the tangential components cancel in pairs.

At the tip, the tension balances the electrostatic and Coulomb forces:

$$T_L \cos\theta_L = qE, \qquad T_L \sin\theta_L = F_C. \tag{20}$$

At the spider, $NT_0 \cos\theta_0 = mg + \zeta_s U$.

### D. Small-angle limit

In the small-angle regime ($\theta \ll 1$), we approximate $\cos\theta \approx 1$ and $\sin\theta \approx \theta$. The tangential tension balance (15) integrates directly to a linear profile $T(s) = T_0 + \eta_\parallel U s$, with total tension variation $T_L - T_0 = \eta_\parallel U L$. We define the tension ratio across the thread:

$$\tau \equiv \frac{T_0}{T_L} = \frac{mg + \zeta_s U}{NqE}. \tag{21}$$

From the first integral (18), the angular profile takes the power-law form $\theta(s) = \theta_L \left[T_L/T(s)\right]^\beta$. Changing the integration variable from arc length $s$ to tension $T$ via $\mathrm{d}s = \mathrm{d}T/(\eta_\parallel U)$, the radial deflection $R = \int_0^L \theta(s)\,\mathrm{d}s$ is evaluated analytically:

$$R = \int_{T_0}^{T_L} \theta_L \left(\frac{T_L}{T}\right)^\beta \frac{\mathrm{d}T}{\eta_\parallel U} = \theta_L L\, g(\tau), \tag{22}$$

where $g(\tau)$ is a geometric shape factor accounting for thread inclination and curvature:

$$g(\tau) \equiv \frac{\tau^{\,1-\beta} - 1}{(\beta - 1)(1 - \tau)}. \tag{23}$$

Note that $\lim_{\tau\to 1} g(\tau) = 1$, recovering a straight, uniformly tilted thread.

Combining $R = \theta_L L g(\tau)$ with the tip boundary condition $\theta_L \approx F_C/(qE) = k_e q S_N/(4R^2 E)$ from Eq. (19), we derive closed-form expressions for **the opening of the bundle**:

$$\boxed{\frac{R}{L} = \left[\frac{S_N\, \Lambda\, g(\tau)}{4}\right]^{1/3}, \qquad \theta_L = \frac{S_N \Lambda}{4}\left(\frac{L}{R}\right)^2} \tag{24}$$

This closed-form solution remains valid for $R \ll L$.

*a. Order of magnitude.* For $N = 2$, $q = 0.6$ nC, $E = 7.41 \times 10^3$ V/m, $L = 0.5$ m, and $\tau \approx 1$: $\Lambda \approx 2.9 \times 10^{-3}$, $R/L \approx 0.09$, $R \approx 4.5$ cm, and $\theta_L \approx 0.09$ rad. The small-angle approximation is therefore justified.

## V. NUMERICAL METHOD

### A. Discretization

The simulations use the discrete elastic rods (DER) method [7] as implemented in [1]. The spider is node $\boldsymbol{x}_0$; each thread $j$ is discretized into $N_t$ edges with nodes $\boldsymbol{x}_{(j-1)N_t+1}, \ldots, \boldsymbol{x}_{jN_t}$, so the system has $NN_t + 1$ nodes and $NN_t$

edges. The first edge of thread $j$ connects the spider to the first node of the thread, and the other edges connect consecutive nodes:

$$\boldsymbol{e}^{(j-1)N_t} = \boldsymbol{x}_{(j-1)N_t+1} - \boldsymbol{x}_0, \qquad \boldsymbol{e}^{(j-1)N_t+m} = \boldsymbol{x}_{(j-1)N_t+m+1} - \boldsymbol{x}_{(j-1)N_t+m}, \quad m = 1, \dots, N_t - 1. \tag{25}$$

Each edge $k = 0, \dots, NN_t - 1$ carries a twist angle $\theta^k$. The degrees of freedom are

$$\boldsymbol{\xi} = \left[\boldsymbol{x}_0, \boldsymbol{x}_1, \dots, \boldsymbol{x}_{NN_t},\ \theta^0, \dots, \theta^{NN_t-1}\right]^T \in \mathbb{R}^{n_{\mathrm{dof}}}, \qquad n_{\mathrm{dof}} = 3(NN_t + 1) + NN_t. \tag{26}$$

The edge vectors depend only on the first $3(NN_t+1)$ components of $\boldsymbol{\xi}$, the node positions; the twist angles enter only the twisting energy. The elastic energy $E_{\mathrm{el}}(\boldsymbol{\xi}) = E_s + E_b + E_t$ contains stretching, bending, and twisting contributions, in the discrete form of [1, 7]. With the edge vectors of Eq. (25), reference edge length $l_0 = L/N_t$, and cross-section $A = \pi r^2$, the stretching energy is

$$E_s = \frac{1}{2} Y A \sum_k \left( \frac{\|\boldsymbol{e}^k\|}{l_0} - 1 \right)^2 l_0. \tag{27}$$

The bending energy $E_b$ (stiffness $YI$, $I = \pi r^4/4$, naturally straight threads) and the twisting energy $E_t$ (stiffness $GJ$, $J = \pi r^4/2$, $G = Y/[2(1+\nu)]$) are those of [7]; for these threads both are negligible ($\overline{YI} \sim 10^{-10}$ [1]). The bending and twisting stencils do not connect different threads through the spider node, which is a free junction. The stretching stiffness is chosen large to enforce inextensibility. The external force vector is

$$\boldsymbol{f}^{\mathrm{ext}}(\boldsymbol{\xi}, \dot{\boldsymbol{\xi}}) = \boldsymbol{W} + \boldsymbol{F}_v + \boldsymbol{F}_r + \boldsymbol{F}_l, \tag{28}$$

with, at node $k$ with Voronoi length $\Delta l_k$ and node tangent $\boldsymbol{t}_k$:

$$\boldsymbol{F}_{v,k} = -\left[\eta_{\parallel}\, \boldsymbol{t}_k \boldsymbol{t}_k^T + \eta_{\perp}(\mathbb{I} - \boldsymbol{t}_k \boldsymbol{t}_k^T)\right] \Delta l_k \left(\boldsymbol{v}_k - \boldsymbol{u}(\boldsymbol{x}_k, t)\right), \tag{29}$$

$$\boldsymbol{F}_{l,k} = q_k\, E(z_k)\, \hat{\boldsymbol{z}}, \tag{30}$$

and the Stokes drag at the spider node. The nodal charges are $q_k = q$ at the tip nodes, zero at the other thread nodes, and $q_0 = Q_s$. The Coulomb force on node $k$ is

$$\boldsymbol{F}_{r,k} = k_e\, q_k \sum_{i \neq k,\ i \neq 0} q_i\, \frac{\boldsymbol{x}_k - \boldsymbol{x}_i}{\|\boldsymbol{x}_k - \boldsymbol{x}_i\|^3}. \tag{31}$$

Since only the tips and the spider carry charge, $\boldsymbol{F}_{r,k}$ vanishes at all interior thread nodes; at the tip nodes it reduces to Eq. (5), and at the spider node it is the force exerted by the $N$ tip charges on $Q_s$. As in [1], the charge of the spider does not act on the thread nodes (sum over $i \neq 0$), while the spider still feels the thread charges; including it changes the forces by $\lesssim Q_s/q \sim 10^{-3}$. In the turbulence ensembles $Q_s = 0$, so $\boldsymbol{F}_r$ acts only between the tips.

### B. Time integration

The equations of motion $m_i \ddot{\xi}_i + \partial E_{\mathrm{el}}/\partial \xi_i - f_i^{\mathrm{ext}} = 0$ are integrated with the implicit (backward) Euler method. At each step we solve the nonlinear residual

$$\boldsymbol{F}(\boldsymbol{\xi}^{n+1}) \equiv \frac{\mathbf{M}}{\Delta t} \left( \frac{\boldsymbol{\xi}^{n+1} - \boldsymbol{\xi}^n}{\Delta t} - \dot{\boldsymbol{\xi}}^n \right) + \nabla E_{\mathrm{el}}(\boldsymbol{\xi}^{n+1}) - \boldsymbol{f}^{\mathrm{ext}}(\boldsymbol{\xi}^{n+1}) = \mathbf{0} \tag{32}$$

by Newton–Raphson, with Jacobian $\mathbf{J} = \mathbf{M}/\Delta t^2 + \nabla^2 E_{\mathrm{el}} - \partial \boldsymbol{f}^{\mathrm{ext}}/\partial \boldsymbol{\xi}$. The drag term is linear in the velocity $(\boldsymbol{\xi}^{n+1} - \boldsymbol{\xi}^n)/\Delta t$, so its contribution to $\mathbf{J}$ is $\mathbf{C}/\Delta t$, with $\mathbf{C}$ the block-diagonal RFT resistance matrix (the dependence of $\boldsymbol{t}_k$ on $\boldsymbol{\xi}$ may be lagged). The Coulomb Jacobian couples the tip nodes and is computed analytically from Eq. (5). The step size is adaptive, as in [1]: divided by 10 on Newton failure, multiplied by 10 after 10 successful steps, and capped at $\Delta t_{\max}$. The rotational inertia of the twist degrees of freedom is $\rho_t J\, l_0/2$ per edge, with Poisson ratio $\nu = 0.5$. A run is aborted if $\Delta t$ falls below $10^{-9}$ s. The default numerical parameters are listed in Table XII. For the isotropic, naturally straight threads considered here, the twist angles stay below $10^{-8}$ rad, as expected for free ends. The procedure is summarized in Algorithm 1; Algorithm 2 describes the parameter sweep used for the phase diagram.

**Algorithm 1** Implicit DER time stepping for a multi-thread ballooning spider in an air flow

**Require:** parameters $\mathcal{P} = \{N, N_t, L, r, m, r_s, q, \text{charge model}, Y, \mu\}$, field $E(z)$, air velocity $\boldsymbol{u}(\boldsymbol{x}, t)$
**Require:** initial state $\boldsymbol{\xi}^0$ (threads vertical, spaced $d_0 = 100$ $\mu$m), $\dot{\boldsymbol{\xi}}^0 = \mathbf{0}$
**Require:** $\Delta t_0$, $\Delta t_{\max}$, tolerance $\epsilon$, max. Newton iterations $K$, final time $t_{\text{end}}$
**Ensure:** trajectory $\{\boldsymbol{\xi}^n, t^n\}$, steady-state flag
$n \leftarrow 0$, $t \leftarrow 0$, $\Delta t \leftarrow \Delta t_0$, $c \leftarrow 0$ ▷ $c$: consecutive successful steps
compute lumped mass matrix $\mathbf{M}$ (spider mass at node 0, thread mass on other nodes)
**while** $t < t_{\text{end}}$ **do**
  $\boldsymbol{\xi}^{(0)} \leftarrow \boldsymbol{\xi}^n$; $k \leftarrow 0$ ▷ Newton initial guess
  **repeat**
    update reference frames by parallel transport; compute material frames from $\boldsymbol{\xi}^{(k)}$
    $\boldsymbol{v}^{(k)} \leftarrow (\boldsymbol{\xi}^{(k)} - \boldsymbol{\xi}^n)/\Delta t$
    $\boldsymbol{f}^{\text{ext}} \leftarrow \boldsymbol{W} + \boldsymbol{F}_v(\boldsymbol{\xi}^{(k)}, \boldsymbol{v}^{(k)}, \boldsymbol{u}) + \boldsymbol{F}_r(\boldsymbol{\xi}^{(k)}) + \boldsymbol{F}_l(\boldsymbol{\xi}^{(k)})$ ▷ Eqs. (29), (31)
    assemble residual $\boldsymbol{F}$ (Eq. (32)) and Jacobian $\mathbf{J}$
    solve $\mathbf{J}\,\Delta\boldsymbol{\xi} = \boldsymbol{F}$; $\boldsymbol{\xi}^{(k+1)} \leftarrow \boldsymbol{\xi}^{(k)} - \Delta\boldsymbol{\xi}$; $k \leftarrow k+1$
  **until** $\|\boldsymbol{F}\|_1 < \epsilon$ **or** $k = K$
  **if** $\|\boldsymbol{F}\|_1 \geq \epsilon$ **then** ▷ Newton failed
    $\Delta t \leftarrow \Delta t/10$; $c \leftarrow 0$ **and retry the step**
  **else**
    $\boldsymbol{\xi}^{n+1} \leftarrow \boldsymbol{\xi}^{(k)}$; $\dot{\boldsymbol{\xi}}^{n+1} \leftarrow (\boldsymbol{\xi}^{n+1} - \boldsymbol{\xi}^n)/\Delta t$; $t \leftarrow t + \Delta t$; $n \leftarrow n+1$
    $c \leftarrow c+1$; **if** $c = 10$ **then** $\Delta t \leftarrow \min(10\,\Delta t, \Delta t_{\max})$, $c \leftarrow 0$
    **save** $(t, \boldsymbol{\xi}^n)$ at fixed output interval ▷ HDF5, for the ML part
    **if** $|\dot{z}_0^n - \dot{z}_0^{n-1}|/|\dot{z}_0^n| < \delta$ over a window $t_w$ **then** ▷ steady state
      **return** trajectory, steady-state flag = true
    **end if**
  **end if**
**end while**
**return** trajectory, steady-state flag = false

**Algorithm 2** Ensemble estimate of the take-off probability

**Require:** grids $\{N\}$, $\{\sigma_w\}$, $\{x\}$ (values in Sec. VII C); $\ell$, $m$, $E$; release height $z_0$, target height $h$; $M(x)$ realizations per point
**for** each $(N, \sigma_w, x)$ **do**
  $w_s \leftarrow mg/(N\eta_\parallel L + \zeta_s)$; $K \leftarrow \sigma_w \ell$; $\bar{F}_l \leftarrow 1 + xK/(w_s h)$; $q \leftarrow \bar{F}_l\, mg/(NE)$ ▷ Eq. (10)
  $n_{\text{up}} \leftarrow 0$
  **for** $i = 1, \ldots, M$ **do**
    draw seed $s_i$; build the modes $\{\boldsymbol{k}_n, \boldsymbol{a}_n, \boldsymbol{b}_n, \omega_n\}$ of Eq. (11)
    hold the spider node fixed at $z_0$ in still air for a time $t_c$ (Algorithm 1 with $\boldsymbol{\xi}_0$ clamped); release at $t = 0$ with $\boldsymbol{u}(\boldsymbol{x}, t)$ switched on
    run Algorithm 1 with $\boldsymbol{u}(\boldsymbol{x}, t)$ from Eq. (11), stopping when $z_0(t) \geq h$, $z_0(t) \leq 0$, or $t \geq t_{\text{end}}$
    **if** $z_0 \geq h$ **then** $n_{\text{up}} \leftarrow n_{\text{up}} + 1$
    record: outcome, exit time, time series of $R(t)$, $\theta_L(t)$, $d_{\min}(t)$; flag ENTANGLED if $d_{\min} < 2r$ at exit
  **end for**
  $\hat{P} \leftarrow n_{\text{up}}/M$ with Wilson 95% interval; compare with Eq. (14)
**end for**

## VI. LEARNED SURROGATE MODEL

### A. Data set

The training data are the 8200 valid trajectories of the sweep of Sec. VII C (Algorithm 2), with $N_t = 50$ edges per thread, i.e. $n = NN_t + 1$ nodes (51 for $N = 1$, 401 for $N = 8$). Each run is stored as one HDF5 file in the layout of Table IV, with frames saved every $\Delta t_{\text{out}} = 0.01$ s from release until exit or $t_{\text{end}} = 60$ s; the frame sequence is contiguous in all 8200 runs. A frame $t$ is a training sample if the $C = 5$ preceding frames exist, so that a velocity

history can be formed. Positions are stored relative to the spider in single precision, and the absolute position of the spider in double precision, so that differences of nearby positions do not lose significant digits.

The data are split by run, never by frame, so that frames of one trajectory do not appear in two splits (Table V). Runs with $N \in \{1, 2, 4\}$ and $\sigma_w \in \{0.15, 0.30\}$ m/s are divided into training, validation, and interpolation test sets in the proportions 80/10/10 at every grid point. Three test sets probe generalization: $N = 3$ (unseen $N$, interpolation), $N = 6$ and 8 (unseen $N$, extrapolation), and $\sigma_w = 0.22$ m/s (unseen turbulence intensity). Means and standard deviations used to normalize inputs and targets are computed on the training set only.

TABLE IV. HDF5 layout: one file per run. Per-frame arrays are stored in single precision; $n_t$ is the number of saved frames and $n = NN_t + 1$ the number of nodes.

| Dataset / attribute | Shape | Content |
|---|---|---|
| `/params` (attributes) | — | $N$, $N_t$, $L$, $m$, $q$, $E$, $z_0$, $h$, $\sigma_w$, $\ell$, $U_h$; charge, field, and flow models; $\Delta t_{\text{out}}$, $t_{\text{end}}$, $\Delta t_{\max}$, code version |
| `/turb` (attributes) | — | seed, $\sigma_w$, $\ell$, $U_h$, $N_k$, renormalization factor, $k_{\min}$, $k_{\max}$ |
| `/turb/k_n`, `a_n`, `b_n`; `omega_n` | $(N_k, 3)$; $(N_k)$ | modes of Eq. (11) |
| `/t` | $(n_t)$ | output times (s) |
| `/x`, `/v` | $(n_t, n, 3)$ | node positions (m) and velocities (m/s); node 0 = spider |
| `/u_air` | $(n_t, n, 3)$ | air velocity $\boldsymbol{u}(\boldsymbol{x}_k, t)$ at the nodes (m/s) |
| `/twist` | $(n_t, NN_t)$ | edge twist angles (rad) |
| `/charge` | $(n)$ | nodal charges (C) |
| `/topology/thread_id` | $(n)$ | thread index per node ($-1$ for the spider) |
| `/topology/node_type` | $(n)$ | spider, interior, or tip |
| `/topology/edges` | $(NN_t, 2)$ | material edges |
| `/outcome` (attributes) | — | rise/fall/timeout, exit time, entangled and steady-state flags; $V$, $R$, $\theta_L$, solver diagnostics |

TABLE V. Data splits. Generalization is tested on numbers of threads and parameter values not seen during training. The range of $q$ covers all $N$ in a split; for each $N$, $q$ follows from the grid in $x$ of Sec. VII C. Frames are saved every 0.01 s.

| Split | $N$ | $q$ (nC) | $\sigma_w$ (m/s) | Runs | Frames |
|---|---|---|---|---|---|
| Train | 1, 2, 4 | 0.16–1.69 | 0.15, 0.30 | 4720 | 4 237 422 |
| Validation | 1, 2, 4 | 0.16–1.69 | 0.15, 0.30 | 590 | 527 828 |
| Test (interpolation) | 1, 2, 4 | 0.16–1.69 | 0.15, 0.30 | 590 | 551 344 |
| Test (unseen $N$, interpolation) | 3 | 0.35–0.53 | 0.30 | 150 | 109 222 |
| Test (unseen $N$, extrapolation) | 6, 8 | 0.08–0.51 | 0.15, 0.30 | 1850 | 1 909 186 |
| Test (unseen $\sigma_w$) | 2, 4 | 0.27–0.73 | 0.22 | 300 | 308 905 |

### B. Graph representation

Each node of the discretized bundle is a graph node. Its input features are: the last $C = 5$ velocities $\boldsymbol{v}_k^{t-C+1}, \ldots, \boldsymbol{v}_k^t$; the air velocity at the node, $\boldsymbol{u}(\boldsymbol{x}_k^t, t)$ and $\boldsymbol{u}(\boldsymbol{x}_k^t, t + \Delta t)$; a learned embedding (dimension 16) of the node type (spider, interior, tip); the nodal charge $q_k$; and the global parameters $qE$, $m$, and $N$, repeated on every node. Absolute positions are not used, so the model is invariant to translations, as in [20]. The second air velocity is evaluated at the *current* positions and the next time. The implicit scheme of Eq. (32) uses the drag at $t + \Delta t$, so this input carries information that the simulator uses; it is available during a rollout without knowledge of $\boldsymbol{x}^{t+1}$. The stored value $\boldsymbol{u}(\boldsymbol{x}_k^{t+1}, t + \Delta t)$ is *not* used, because it depends on the target.

Two sets of edges are used. *Material edges* connect consecutive nodes of each thread and the spider to the first node of each thread, in both directions; their features are the relative position $\boldsymbol{x}_i - \boldsymbol{x}_j$, the distance $|\boldsymbol{x}_i - \boldsymbol{x}_j|$, and the stretch $|\boldsymbol{x}_i - \boldsymbol{x}_j|/l_0$. *Interaction edges* connect nodes of different threads: all pairs closer than $R_c = 5$ cm, which resolves the proximity of the threads near the spider, and all pairs of tips, which carry the Coulomb repulsion of the

tip charges [Eq. (5)] at any distance. Their features are $\boldsymbol{x}_i - \boldsymbol{x}_j$, $|\boldsymbol{x}_i - \boldsymbol{x}_j|$, $1/|\boldsymbol{x}_i - \boldsymbol{x}_j|^2$ (clipped), $q_i q_j$, and a flag for the edge type. The spider is excluded from the radius graph, since it is connected to every thread by material edges. For $N = 1$, 3, 4, 6, and 8, a graph has 100, 300, 400, 600, and 800 material edges and 0, 2346, 3806, 6714, and 9621 interaction edges. Fully connecting the tips costs $N(N-1)$ edges, which is negligible for $N \leq 8$; for much larger bundles, long-range schemes such as Ewald message passing [15] would replace it.

### C. Architecture

The surrogate is a graph network simulator in the encoder–processor–decoder form of [16, 20], within the graph network framework of [17]. The encoder maps the node features $\boldsymbol{f}_i$ and the edge features $\boldsymbol{r}_{ij}$ of each edge set $s$ to latent vectors of size $L_h = 128$,

$$\boldsymbol{h}_i^0 = \epsilon^v(\boldsymbol{f}_i), \qquad \boldsymbol{e}_{ij}^{s,0} = \epsilon^s(\boldsymbol{r}_{ij}). \tag{33}$$

The processor applies $M = 10$ message-passing steps with residual updates and separate weights at every step,

$$\boldsymbol{e}_{ij}^{s,m+1} = \boldsymbol{e}_{ij}^{s,m} + \phi^{s,m}\left(\boldsymbol{e}_{ij}^{s,m}, \boldsymbol{h}_i^m, \boldsymbol{h}_j^m\right), \tag{34}$$

$$\boldsymbol{h}_i^{m+1} = \boldsymbol{h}_i^m + \gamma^m\Big(\boldsymbol{h}_i^m, \textstyle\sum_{j\in\mathcal{N}_{\mathrm{mat}}(i)} \boldsymbol{e}_{ij}^{\mathrm{mat},m+1}, \sum_{j\in\mathcal{N}_{\mathrm{int}}(i)} \boldsymbol{e}_{ij}^{\mathrm{int},m+1}\Big), \tag{35}$$

with one sum per edge set. The decoder gives the normalized acceleration of every node, $\hat{\boldsymbol{a}}_i = \delta(\boldsymbol{h}_i^M)$. All functions $\epsilon$, $\phi$, $\gamma$, $\delta$ are multilayer perceptrons with two hidden layers of width 128 and ReLU activations; their outputs are layer-normalized, except for the decoder. Ten steps propagate information over ten material edges, i.e. one fifth of a thread; the interaction edges connect the tips directly. The model is implemented in PyTorch and PyTorch Geometric [18].

We train two configurations: *GNS-local*, with material edges only and without the air velocity at $t + \Delta t$ ($1.59 \times 10^6$ parameters), the closest to the original GNS; and *GNS*, with both edge sets and all inputs ($2.62 \times 10^6$ parameters).

*a. Target and integration.* The target is the acceleration $\boldsymbol{a}_k^t = (\boldsymbol{v}_k^{t+1} - \boldsymbol{v}_k^t)/\Delta t$, computed from the stored velocities, normalized with the training mean and standard deviation of each component. The state is advanced by the semi-implicit Euler update

$$\boldsymbol{v}_k^{t+1} = \boldsymbol{v}_k^t + \Delta t\, \boldsymbol{a}_k^t, \qquad \boldsymbol{x}_k^{t+1} = \boldsymbol{x}_k^t + \Delta t\, \boldsymbol{v}_k^{t+1}, \tag{36}$$

which reproduces the stored velocities exactly and the stored positions to $1.2 \times 10^{-7}$ m, the round-off level of the single-precision storage.

*b. Training.* The loss is the mean squared error of the normalized acceleration over all nodes and components. To make the model robust to its own errors during rollouts, the velocity history is perturbed by a random walk [20], scaled so that the standard deviation of the last velocity is $k\,\sigma_{a,c}^{(\tau)}\Delta t$, where $\sigma_{a,c}^{(\tau)}$ is the training standard deviation of acceleration component $c$ for node type $\tau$; the target is corrected accordingly, $\boldsymbol{a}_k^t = (\boldsymbol{v}_k^{t+1} - \tilde{\boldsymbol{v}}_k^t)/\Delta t$. The noise is set per node type because the accelerations of the spider, which has most of the inertia, are several times smaller than those of the threads. We train with $k \in \{0.1, 0.3, 1, 3\}$ and keep $k = 1$, which gives both the longest physical rollouts and the smallest position error at 5 s on the 32 validation runs. Optimization uses Adam [19], with a learning rate decaying exponentially from $10^{-4}$ to $10^{-6}$ over 15 epochs, batches of 128 (GNS-local) and 64 (GNS) graphs, and early stopping when the rollout error on 32 validation runs, stratified by $N$ and $\sigma_w$, has not improved for two epochs. Training runs on one NVIDIA L4 GPU (24 GB): an epoch, including the validation rollouts, takes about 0.8 h for GNS-local and 3.0 h for GNS. GNS-local was trained for 14 epochs (11.6 h) and GNS for 10 epochs (30.5 h), when the compute budget was reached; the checkpoint used is the one with the smallest validation error at 5 s. The procedure is summarized in Algorithm 3.

*c. Rollouts.* A rollout starts from the first $C$ simulated frames after release and then uses only its own predictions (Algorithm 4). At every step the graph is rebuilt from the predicted positions and the air velocity is evaluated from the modes of the run, with the same formula as the simulator [Eq. (11)]. A rollout ends with the same criteria as the simulation (Algorithm 2): $z_0 \geq h$, $z_0 \leq 0$, or $t \geq t_{\mathrm{end}}$.

**Algorithm 3** Training of the graph network simulator

**Require:** training runs; history length $C$; noise factor $k$; training statistics (means, standard deviations of inputs and targets, $\sigma_{a,c}^{(\tau)}$)
**Ensure:** trained parameters $\Theta$
**for** each training step **do**
  sample a batch of frames $(r, t)$ from the training runs
  **for** each frame **do**
    read $\boldsymbol{v}^{t-C+1..t}$, $\boldsymbol{v}^{t+1}$, $\boldsymbol{x}^t$ (relative to the spider), $\boldsymbol{u}(\boldsymbol{x}^t, t)$, $\boldsymbol{u}(\boldsymbol{x}^t, t+\Delta t)$, node types, charges, globals
    add the random walk to $\boldsymbol{v}^{t-C+1..t}$, with standard deviation $k\,\sigma_{a,c}^{(\tau)}\Delta t$ at the last velocity
    target $\boldsymbol{a}^t \leftarrow (\boldsymbol{v}^{t+1} - \tilde{\boldsymbol{v}}^t)/\Delta t$, normalized
    build the material and interaction edges from $\boldsymbol{x}^t$; compute node and edge features
  **end for**
  $\hat{\boldsymbol{a}} \leftarrow$ encoder–processor–decoder$_\Theta$(batch)
  loss $\leftarrow$ mean of $|\hat{\boldsymbol{a}} - \boldsymbol{a}|^2$ over nodes and components; update $\Theta$ with Adam
  every $10^3$ steps: one-step loss on fixed validation frames; periodically: rollouts on 32 validation runs
**end for**

**Algorithm 4** Rollout of the graph network simulator for one run

**Require:** trained $\Theta$; first $C$ frames of the run after release; modes $\{\boldsymbol{k}_n, \boldsymbol{a}_n, \boldsymbol{b}_n, \omega_n\}$ of the run; $h$, $t_{\text{end}}$
**Ensure:** predicted trajectory and outcome (rise, fall, timeout)
$t \leftarrow t_C$; initialize $\boldsymbol{x}$, $\boldsymbol{v}^{t-C+1..t}$ from the simulated frames
**while** $t < t_{\text{end}}$ **do**
  $\boldsymbol{u}^t \leftarrow \boldsymbol{u}(\boldsymbol{x}^t, t)$, $\boldsymbol{u}^{t+\Delta t} \leftarrow \boldsymbol{u}(\boldsymbol{x}^t, t+\Delta t)$ from Eq. (11), at the current absolute positions
  build the graph from $\boldsymbol{x}$; $\hat{\boldsymbol{a}} \leftarrow$ encoder–processor–decoder$_\Theta$; de-normalize
  update $\boldsymbol{v}$, $\boldsymbol{x}$ with Eq. (36); spider position in double precision; thread positions kept relative to the spider, $\boldsymbol{x}_k - \boldsymbol{x}_0$; shift the velocity history
  $t \leftarrow t + \Delta t$
  **if** $z_0 \geq h$ **then return** rise; **if** $z_0 \leq 0$ **then return** fall
**end while**
**return** timeout

### D. Reference methods

Two methods without learning set the floor that the surrogate has to beat: *constant velocity*, in which every node keeps its last velocity, and *follow-air*, in which every node, the spider included, is advected by the air velocity at its current absolute position and at the current time, $\boldsymbol{v}_k = \boldsymbol{u}(\boldsymbol{x}_k, t)$, as a massless tracer. The second is a strong reference over short times, because the threads have almost no inertia and follow the air closely.

### E. Evaluation metrics

All methods are evaluated on the same test runs and frames:

- one-step error of the normalized acceleration, on a fixed random sample of $10^5$ frames per test set;
- rollout position error at 0.2, 1, 5, and 10 s after release, separately for the spider (absolute position) and for the thread nodes (relative to the spider), and the fraction of rollouts that remain physical (finite, with segment lengths within 10% of $l_0$);
- take-off probability: $P$ from the surrogate rollouts at every grid point $(N, \sigma_w, x)$, compared with the simulator, $|\Delta P|$ (mean and maximum per test set);
- computational cost: wall-clock time per simulated second and per estimate of $P$, one-time costs of data generation and training, break-even number of evaluations (39), and error of $P$ at equal compute budget (Sec. VII D 1).

## VII. RESULTS

We present the numerical and analytical results in four parts. First, Sec. VII A validates our implicit DER numerical solver against exact analytical benchmarks for terminal velocity and reproduces established multi-thread cases from the literature [1]. Second, Sec. VII B compares multi-thread steady-state geometries against the analytical opening law and first integral derived in Sec. IV. Third, Sec. VII C evaluates take-off probabilities in turbulent air relative to the drift–diffusion model. Fourth, Sec. VII D assesses the generalization capability of the graph neural network surrogate.

### A. Validation of the numerical model

#### 1. Analytical check: terminal velocity

From Eq. (7), for a vertical thread in still air and $\zeta_s \ll N\eta_\parallel L$:

$$U \approx \frac{NqE - mg}{N\eta_\parallel L} = \frac{\ln(L/r) - \frac{1}{2}}{2\pi}\,\frac{NqE - mg}{N\mu L} \approx 2.2\,\frac{NqE - mg}{N\mu L}. \tag{37}$$

Habchi and Jawed normalize the terminal velocity by $v_{\text{charac}} = (NqE - mg)/(\mu NL)$ and report a numerical steady value $\bar{v}_t \approx 2.2$ for a single thread [1]. Equation (37) explains this constant analytically: it corresponds precisely to the geometric factor $[\ln(L/r) - 1/2]/(2\pi)$ stemming from slender-body RFT theory. Including the Stokes drag of the spider body, the exact dimensionless terminal velocity for straight vertical threads becomes:

$$\bar{v}_t^{\,\text{st}} = \frac{\mu NL}{N\eta_\parallel L + \zeta_s}, \tag{38}$$

which reduces to Eq. (37) for $\zeta_s \ll N\eta_\parallel L$. The correction is not small for short threads: $\zeta_s/(\eta_\parallel L) = 0.37$ for $L = 0.1$ m and 0.08 for $L = 0.5$ m. For $N > 1$ the threads tilt, the normal component $\eta_\perp > \eta_\parallel$ enters the drag, and $\bar{v}_t$ falls below Eq. (38).

Table VI compares both expressions with the simulations. The numerical values agree with Eq. (38) within 0.36–0.45%, consistent with the discretization error at $N_t = 100$ (Table VIII); Eq. (37) overestimates $\bar{v}_t$ by up to 36% for $L = 0.1$ m, because it neglects the body drag.

TABLE VI. Dimensionless terminal velocity $\bar{v}_t = U\mu NL/(NqE - mg)$ for a single vertical thread ($N = 1$, $r = 300$ nm, $m = 1$ mg, $Q_s = 0$, tip charge with $\bar{F}_l = 2$, still air, constant field $E = 7.41 \times 10^3$ V/m, $N_t = 100$). Analytical values from Eq. (37) (thread drag only) and Eq. (38) (with the Stokes drag of the body); numerical values from Algorithm 1 at steady state. The relative error is with respect to Eq. (38).

| $L$ (m) | $\ln(L/r)$ | $\bar{v}_t$, Eq. (37) | $\bar{v}_t$, Eq. (38) | $\bar{v}_t$ numerical | Relative error (%) |
|---|---|---|---|---|---|
| 0.1 | 12.72 | 1.944 | 1.423 | 1.428 | 0.36 |
| 0.5 | 14.33 | 2.201 | 2.032 | 2.041 | 0.45 |
| 1.0 | 15.02 | 2.311 | 2.214 | 2.224 | 0.45 |

#### 2. Reproduction of the validation in [1]

1. **Test 1.** One thread of 0.5 m in the chamber field of Eq. (1), with the mass of the spiders of Morley and Gorham, $m = 0.9$ mg [3], and $Q_s = 0$. Habchi and Jawed report a steady velocity of $\approx 8.5$ cm/s reached after $\approx 0.1$ s [1], but do not state the charge or its distribution for this case. With tip charge, the tip sits at $z \approx L$, where $E \simeq E_\infty = 7.41 \times 10^3$ V/m, so the steady velocity follows from Eq. (7); we choose $q = (mg + U\zeta_{\text{tot}})/E_\infty = 1.2433$ nC, with $\zeta_{\text{tot}} = \eta_\parallel L + \zeta_s$, to target $U = 8.5$ cm/s. This test therefore checks the force balance and the transient, not the value 8.5 cm/s independently. We run two variants: (a) the charge at the tip, and (b) the same charge on the spider node, which mimics the one-dimensional model of [3] and places the charge in the enhanced field near the prominence.

2. **Test 2.** One point of Fig. 6 in [1]: $N = 2$, tip charge, $\bar{F}_l = 2$ ($q = 1.324$ nC), $m = 1$ mg, $Q_s = 3$ pC, constant field $E = 7.41 \times 10^3$ V/m. Target: $\bar{v}_t \approx 2$, read from the figure.

3. **Test 3.** Discretization convergence for test 2: $N_t = 25, 50, 100, 200$ (Table VIII).

4. **Code checks.** Finite-difference verification of all gradients, Hessians, and Jacobians; terminal velocity against Eq. (7), with and without $Q_s$; the first integral (18); and the uniform-flow invariance of Sec. III.

For the uniform-flow check, both runs use a fixed step $\Delta t = 3 \times 10^{-5}$ s and the default Newton tolerance, and the shapes are compared in the spider frame every 0.1 s up to $t = 2$ s. When initialized with identical relative velocities, the thread profiles in the moving and stationary air frames coincide to within $8.5 \times 10^{-12} L$, demonstrating Galilean invariance down to floating-point precision. When both simulations are initialized from rest in the ground frame, the shape mismatch decays exponentially with a characteristic relaxation time constant of $\tau_r \approx 0.24$ s, roughly four times the vertical translational time constant $t_s = m/(N\eta_\parallel L + \zeta_s) = 0.059$ s. The tip radius deviation $R(t)$ relaxes with this same time scale (0.238 s), identifying the slowest physical degree of freedom as the transverse Coulomb opening of the bundle resisted by normal RFT drag.

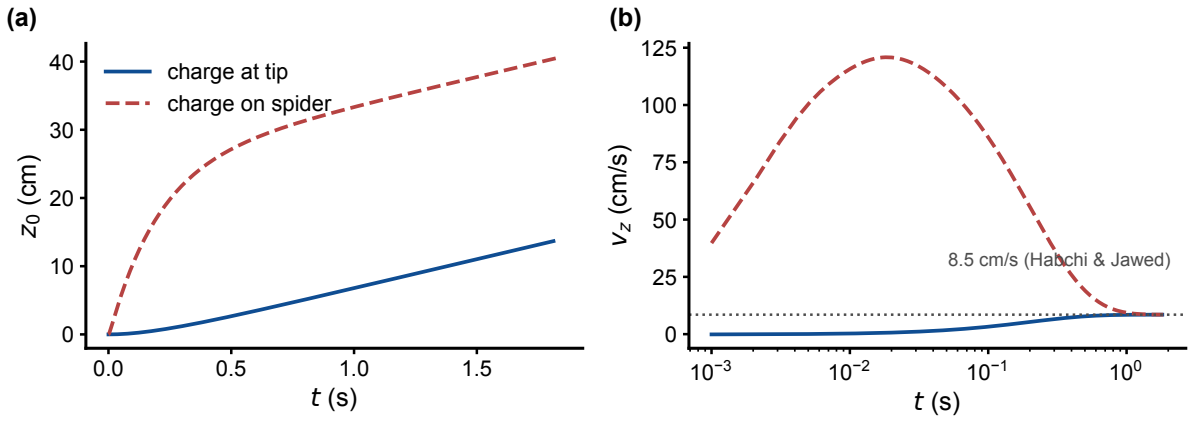



FIG. 2. Test 1: take-off of a spider ($m = 0.9$ mg, $Q_s = 0$) with one thread ($L = 0.5$ m, $N_t = 100$) in the chamber field of Eq. (1), with $q = 1.2433$ nC. (a) Altitude $z_0(t)$ and (b) vertical velocity $v_z(t)$ of the spider node, for the charge at the thread tip (solid) and on the spider node (dashed). The dotted line marks the steady velocity $\approx 8.5$ cm/s reported in [1]. With tip charge the velocity relaxes on the time scale $t_s = m/(\eta_\parallel L + \zeta_s) = 0.20$ s; with the charge on the spider, the enhanced field near the prominence ejects the spider at up to 1.21 m/s before it settles.

### B. Analytical vs. numerical comparison

Table IX compares the steady opening of tip-charged bundles with Eq. (24). The simulated tip radius is 0.8%, 1.3%, and 1.8% below the prediction for $N = 2$, 4, and 8, and the tip angle $\theta_L$ is 1.8–3.7% above it, although $R/L$ reaches 0.20. The deviations grow with $R/L$, as expected from the small-angle approximation. The normalized velocity is 1.4–3.3% below the straight-thread value of Eq. (38), which measures the extra drag due to the tilt of the threads. The threads never come closer than 4 mm, so no entanglement occurs in still air. Figure 3 extends the comparison to $\bar{F}_l = 1.2$–5: the data for all $N$ collapse on the line of slope 1 up to $R/L \approx 0.2$, and fall below it at larger openings, by about 10% at $R/L \approx 0.3$, where the small-angle approximation breaks down.

### C. Take-off in turbulence

*a. Data set.* All runs use $m = 1$ mg, $L = 0.5$ m, a height-independent field $E = 7.41 \times 10^3$ V/m (the far-field value $E_\infty$ of Eq. (1)), $\ell = 1$ m, $U_h = 1$ m/s, $z_0 = 0.5$ m, $h = 2$ m, $N_t = 50$, and $t_{\text{end}} = 60$ s. The sweep of Algorithm 2 covers $N = 1, 2, 4, 8$, $\sigma_w = 0.15$ and 0.30 m/s, and seven values of the scaled drift $x = w_s(\bar{F}_l - 1)h/K$, $x \in \{-4, -2, -1, 0, 1, 2, 4\}$, with $K = \sigma_w \ell$; $M = 200$ realizations for $|x| \le 1$ and $M = 100$ otherwise (8000 realizations), plus $N = 3, 6$ at $\sigma_w = 0.30$ m/s and $N = 2, 4$ at $\sigma_w = 0.22$ m/s for $x \in \{-1, 0, 1\}$, with $M = 50$ (600 realizations).

TABLE VII. Reproduction of the validation cases of [1].

| Test | Configuration | Reference value | This work | Error (%) |
|---|---|---|---|---|
| 1a | $N = 1$, $m = 0.9$ mg, tip charge, Eq. (1) | $U \approx 8.5$ cm/s [1] | 8.502 cm/s | 0.02 |
| 1a | same | $t_{95} = 3t_s = 0.597$ s (theory) | 0.604 s | 1.1 |
| 1a | same | time to steady state $\approx 0.1$ s [1] | 0.604 s ($t_{95}$) | — |
| 1b | same, charge on the spider node | $U \approx 8.5$ cm/s [1] | 8.516 cm/s | 0.19 |
| 1b | same | — | peak 1.21 m/s; within 5% of $U$ after 1.14 s | — |
| 2 | $N = 2$, tip charge, $\bar{F}_l = 2$, $N_t = 100$ | $\bar{v}_t \approx 2$ (read from Fig. 6 of [1]) | 2.088 | 4.4 |
| check | $N = 1$, constant $E$, $Q_s = 0$ | Eq. (7): 9.513 cm/s | 9.520 cm/s | 0.076 |
| check | $N = 1$, constant $E$, $Q_s = 3$ pC | Eq. (7) with $Q_sE$: 10.044 cm/s | 10.050 cm/s | 0.066 |
| check | $N = 2$, first integral (18) | $T^\beta \sin\theta = \text{const.}$ | spread 0.019% | — |
| check | $N = 4$, uniform flow $w = 0.5$ m/s, same initial state relative to the air | $\Delta V = 0.5$ m/s; same shape at all $t$ | $\Delta V = 0.50000000005$ m/s; max. shape deviation $8.5 \times 10^{-12} L$ over $t \le 2$ s | $10^{-8}$ |
| check | same, both runs started at rest in the ground frame | transient, same steady state | deviation $2.3\times10^{-3} L$ at $t = 0.1$ s, $9 \times 10^{-7} L$ at $t = 2$ s; decay time 0.24 s | — |
| check | gradients, Hessians, Jacobians | finite differences | agreement to $\sim 10^{-10}$ | — |

TABLE VIII. Discretization convergence for test 2 ($N = 2$, tip charge, $\bar{F}_l = 2$, $Q_s = 3$ pC). $\Delta$ is the relative difference with respect to the finest mesh, $N_t = 200$. The differences halve with each refinement (first-order convergence); Richardson extrapolation gives $\bar{v}_t \to 2.078$. CPU time: one core of an Apple M4 Pro.

| $N_t$ (edges/thread) | $\bar{v}_t$ | $\Delta\bar{v}_t$ (%) | $R/L$ | $\Delta(R/L)$ (%) | CPU time (s) |
|---|---|---|---|---|---|
| 25 | 2.1196 | 1.75 | 0.14287 | 0.48 | 5.6 |
| 50 | 2.0986 | 0.74 | 0.14248 | 0.20 | 11.5 |
| 100 | 2.0883 | 0.25 | 0.14229 | 0.067 | 25.7 |
| 200 | 2.0832 | — | 0.14220 | — | 51.6 |

The grid is uniform in $x$, not in $\bar{F}_l$: the charge $q$ follows from $x$ through Algorithm 2, so each $(N, \sigma_w)$ has its own set of $\bar{F}_l$ (Table XIII in Appendix D). Four grid points with large $N\sigma_w$ and $x \le -2$ have $\bar{F}_l \le 0$, i.e. no electrostatic lift; they lie outside the model (the threads fold onto the spider) and are excluded (387 realizations; all 13 solver failures of the sweep occurred there). Of the remaining 8200 realizations, 56 (0.7%) reached $t_{\text{end}} = 60$ s without exit and are excluded from $P$; no thread entanglement was detected (in 3 of 8200 realizations the threads approach to 0.5–0.6 $\mu$m, about $2r$, a few centimetres from the spider within 0.15 s after release, and then separate).

*b. Collapse on the scaled drift.* Figure 7(a) shows that the transition from failure to take-off widens with $N$ when plotted against $\bar{F}_l$, because the settling speed $w_s$ decreases with $N$ and the same turbulence then competes with a smaller drift. In the scaled variable $x$, Fig. 7(b), all 64 grid points collapse on a single curve, independently of $N$ and $\sigma_w$. This is the structure predicted by Eq. (14), which depends on the parameters only through $x$. The width of the curve, however, is not the one of Eq. (14) with $K = \sigma_w \ell$: the simulated transition is about four times sharper. A maximum-likelihood fit of Eq. (14) with an effective diffusivity $K_{\text{eff}} = K/a$ and an effective start fraction $p_0$ in place of $z_0/h$ gives $a = 4.2$ and $p_0 = 0.34$, with $\chi^2 = 48$ for 62 degrees of freedom. Separate fits for each $(N, \sigma_w)$ give $a = 3.7$–$4.5$ and $p_0 = 0.32$–$0.38$, without a systematic trend with $N$.

*c. Interpretation.* Both corrections are properties of the flow, not of the bundle. The diffusivity of Taylor's theory is $K = \sigma_w^2 T_L$, with $T_L$ the Lagrangian integral time; $K = \sigma_w \ell$ assumes $T_L = \ell/\sigma_w$, whereas $a = 4.2$ corresponds to $T_L \approx 0.24\,\ell/\sigma_w$. At $x = 0$, a massless tracer in the same fields reaches $h$ with probability 0.33 ($[0.20, 0.48]$, 40 fields), consistent with $p_0 = 0.34$; the excess over $z_0/h = 0.25$ reflects the finite correlation time of the velocity seen during the ascent. A tracer study in the same fields confirms both values: with an imposed drift $U_0 = xK/h$, massless tracers ($M = 2000$ per point) give $a = 4.20$, $p_0 = 0.34$ for $\sigma_w = 0.15$ m/s and $a = 4.12$, $p_0 = 0.36$ for $\sigma_w = 0.30$ m/s,

TABLE IX. Steady-state observables for tip-charged bundles in still air ($L = 0.5$ m, $m = 1$ mg, $Q_s = 0$, $\bar{F}_l = 2$, constant $E = 7.41 \times 10^3$ V/m, $N_t = 100$). $\tau$ uses the simulated $U$. $\Lambda$, $\tau$, and $(R/L)_{\mathrm{an}}$ from Eqs. (23)–(24); the remaining columns from simulation. $d_{\mathrm{min}}$: minimum distance between threads.

| $N$ | $q$ (nC) | $\Lambda$ | $\tau$ | $(R/L)_{\mathrm{an}}$ | $(R/L)_{\mathrm{num}}$ | $\theta_L$ (rad) | $\bar{v}_t$ | $d_{\mathrm{min}}$ ($\mu$m) |
|---|---|---|---|---|---|---|---|---|
| 2 | 1.324 | $6.42 \times 10^{-3}$ | 0.520 | 0.1433 | 0.1422 | 0.0796 | 2.084 | 7497 |
| 4 | 0.662 | $3.21 \times 10^{-3}$ | 0.510 | 0.1790 | 0.1766 | 0.0987 | 2.103 | 6584 |
| 8 | 0.331 | $1.61 \times 10^{-3}$ | 0.505 | 0.2039 | 0.2003 | 0.1123 | 2.106 | 4021 |

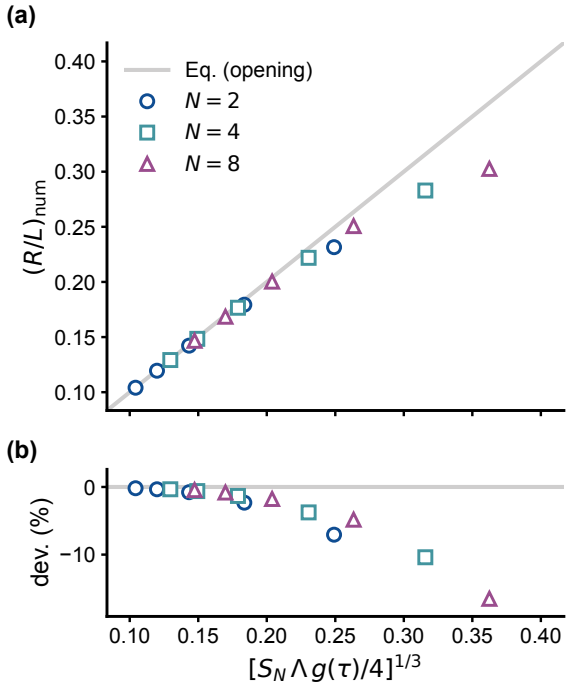



FIG. 3. Opening of the bundle. (a) Tip radius from simulation, $(R/L)_{\mathrm{num}}$, versus the small-angle prediction of Eq. (24), for $N = 2$ (circles), 4 (squares), and 8 (triangles), and $\bar{F}_l = 1.2$, 1.5, 2, 3, 5 (tip charge, still air, $m = 1$ mg, $Q_s = 0$, constant $E = 7.41 \times 10^3$ V/m). The solid line has slope 1. (b) Relative deviation from the prediction: below 1% for small openings, growing to about −17% at the largest opening ($N = 8$, $\bar{F}_l = 5$), which marks the breakdown of the small-angle approximation.

identical within the uncertainties to the fit for the bundles, and the Lagrangian integral time of $w$ along tracer paths gives $a = \sigma_w \ell/(\sigma_w^2 T_L) = 3.0$–$5.8$, depending on the integration limit of the autocorrelation. The bundle therefore enters the take-off probability only through its settling speed $w_s$, i.e. through $x$: within the statistical resolution of the sweep ($\pm 0.07$ in $P$ at the transition), the deformation of the threads and their Coulomb opening have no further effect. Equation (14) remains the correct form, provided $K$ is the Lagrangian diffusivity $\sigma_w^2 T_L$ of the flow and $z_0/h$ is replaced by the effective start fraction of the same flow.

*d. Deformation.* Turbulence opens the bundle on average: for $N = 4$, $\bar{F}_l = 1$, $\sigma_w = 0.30$ m/s, the time-averaged tip radius is $R/L = 0.165 \pm 0.018$ over the realizations, with temporal fluctuations of 0.039 within a run, against 0.115 in still air [Fig. 8(b)]. The mean opening over all realizations grows with $N$: $R/L = 0.042$, 0.112, 0.149, and 0.204 for $N = 1$, 2, 4, and 8 (for $N = 1$, the tilt of the single thread). The closest approach between threads is 0.5 $\mu$m. Clamped in still air for $t \geq 1.4$ s, the bundle reaches $R/L = 0.093$, 0.115, and 0.127 for $N = 2$, 4, and 8, within 0.2, 0.2, and 3.2% of Eq. (24); in turbulence the run-averaged opening is larger by 26, 43, and 83%. The sweep releases the bundle after a clamp time $t_c = 0.05$ s, before the Coulomb opening is complete ($R/L \approx 0.07$ at release). A control run with $t_c = 3$ s, for $N = 4$, 8, $\sigma_w = 0.30$ m/s, $x = 0$, 1, and the same 200 seeds per point, gives take-off probabilities within 0.017 of the sweep (two-proportion test $p \geq 0.73$) and run-averaged openings within 2%: the results above do not depend on the initial opening.

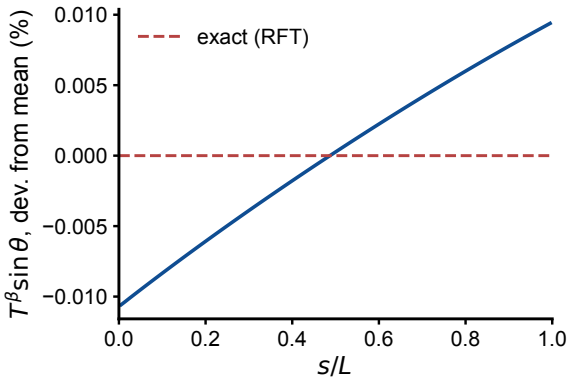



FIG. 4. Numerical check of the first integral of Eq. (18). The quantity $T^\beta \sin\theta$, normalized by its mean value, along one thread at steady state ($N = 2$, tip charge, $q = 1.324$ nC, $\bar{F}_l = 2$, $Q_s = 0$), for $N_t = 100$. Shown is the deviation from the mean in percent; the exact RFT result is zero (dashed line). The spread over the interior edges is 0.019%.

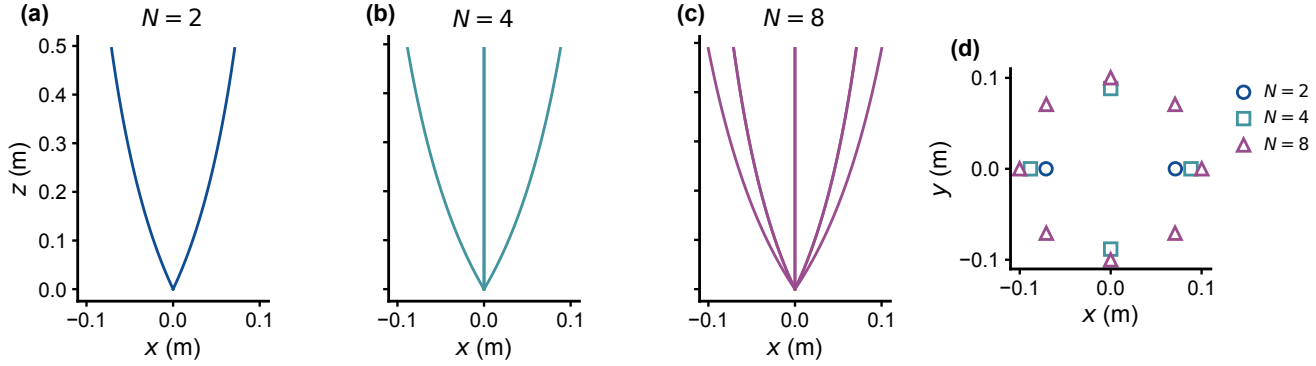



FIG. 5. Steady shapes of tip-charged bundles in still air ($L = 0.5$ m, $m = 1$ mg, $Q_s = 0$, $\bar{F}_l = 2$, constant $E = 7.41 \times 10^3$ V/m, $N_t = 100$; parameters of Table IX). (a)–(c) Side view of the threads in the frame of the spider for $N = 2$, 4, and 8. (d) Top view of the tips, which lie on a circle of radius $R$, as assumed in Sec. IV.

### D. Learned surrogate

Table X compares the two GNS configurations with the reference methods on the four test sets. The interpolation set measures the accuracy within the training distribution, the two unseen-$N$ sets the transfer to bundles with a different number of threads, and the unseen-$\sigma_w$ set the transfer to a different turbulence intensity. Figure 10(a)–(d) shows how the rollout error grows with time, and Fig. 10(e) compares the take-off probabilities predicted by the surrogate with those of the simulator at every grid point. On trajectories shorter than about 1 s, both learned models beat the two references: at 1 s the thread-node error of GNS is 19 mm on the interpolation set, against 157 mm for follow-air and 392 mm for constant velocity. The advantage does not carry over to the take-off probability. $|\Delta P|$ on the interpolation set is 0.14 for constant velocity and 0.45–0.46 for the two GNS configurations, and at some grid points the GNS outcome disagrees with the simulator in every run ($|\Delta P| = 1$). The cause is the length of the rollouts. The median time until a segment length leaves the 10% tolerance is 0.75 s for GNS and 0.97 s for GNS-local on the interpolation set, and 0.05 s for both on the unseen $N = 6$ and 8, while a flight lasts from a few seconds to tens of seconds. Follow-air, which carries neither weight nor electrostatic force, gives a take-off probability between 0.2 and 0.5 at every grid point and hence a large $|\Delta P|$; that is the behaviour expected of a massless tracer, not a failure of the method. Neither learned model transfers to bundles with more threads than it was trained on: the one-step error rises from 0.12 (GNS) and 0.087 (GNS-local) on the interpolation set to 4.5 and 18 on $N = 6$ and 8.

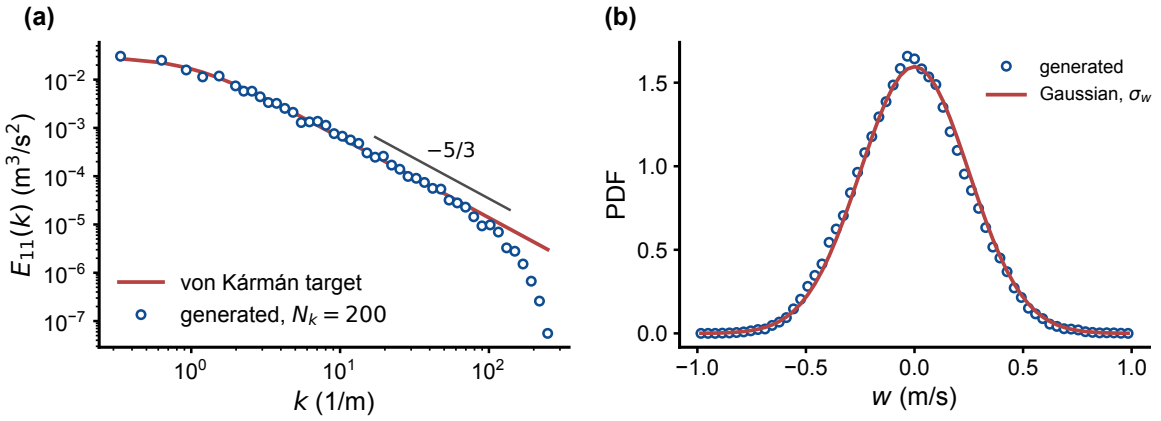



FIG. 6. Validation of the synthetic turbulence of Eq. (11), for $\sigma_w = 0.25$ m/s, $\ell = 1$ m, and $N_k = 200$ modes (average over 50 realizations). (a) Longitudinal one-dimensional spectrum of $w$ along a vertical line versus the spectrum implied by Eq. (12); the generated spectrum is shown up to the cut-off $k_{\max} = 2\pi/(5 \times 5 \text{ mm}) = 251 \text{ m}^{-1}$ and falls below the target above $k \approx 80 \text{ m}^{-1}$ because of the finite shell width near $k_{\max}$. (b) Probability density of $w$ with the Gaussian of standard deviation $\sigma_w$; the sample standard deviation is 0.249 m/s. Mode amplitudes renormalized as in Appendix E.

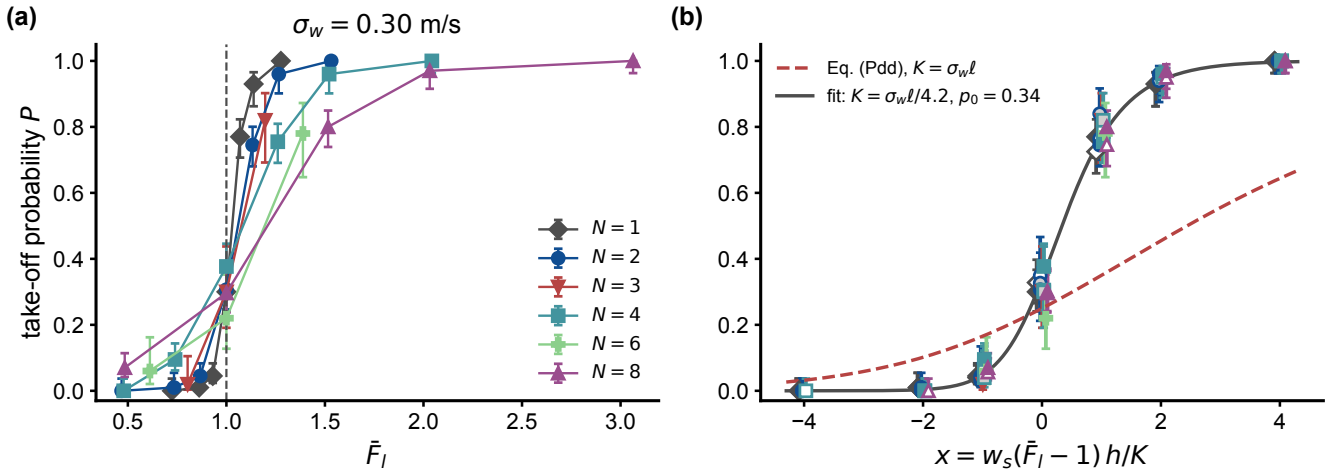



FIG. 7. Take-off probability in synthetic turbulence ($\ell = 1$ m, $U_h = 1$ m/s, $z_0 = 0.5$ m, $h = 2$ m, tip charge, $m = 1$ mg, $Q_s = 0$, $N_t = 50$). Markers: ensemble estimates from Algorithm 2 with 95% Wilson intervals ($M = 200$ for $|x| \le 1$, $M = 100$ otherwise; $M = 50$ for $N = 3$, 6 and $\sigma_w = 0.22$ m/s). (a) $P$ versus $\bar{F}_l$ for $\sigma_w = 0.30$ m/s; the vertical dashed line marks the deterministic threshold $\bar{F}_l = 1$. (b) All valid grid points versus the scaled drift $x = w_s(\bar{F}_l - 1)h/K$, $K = \sigma_w \ell$; open markers $\sigma_w = 0.15$, grey 0.22, filled 0.30 m/s, small horizontal offsets for readability. Dashed: Eq. (14) with $K = \sigma_w \ell$; solid: Eq. (14) with $K_{\text{eff}} = \sigma_w \ell / a$ and start fraction $p_0$ fitted to all points ($a = 4.2$, $p_0 = 0.34$).

### 1. *Computational cost*

*a. Protocol.* The surrogate and the simulator are compared on the same tasks: (i) one complete run, from release until the spider reaches $h$ or the ground, and (ii) one estimate of the take-off probability $P$ from $M$ realizations. Times are the wall-clock times measured during the test evaluation of Sec. VII D, on the interpolation test set. The simulator runs one realization per vCPU of a Google Cloud `c2-standard-30` instance (30 vCPUs); the surrogate runs on one NVIDIA L4 GPU (24 GB) of a `g2-standard-8` instance (8 vCPUs, 32 GB), and its time includes the evaluation of the air velocity [Eq. (11)] at every step and the construction of the graph. Because runs end at different times, the speed-up of Table X is defined per simulated second: the simulator wall-clock time per simulated second divided by that of the surrogate.

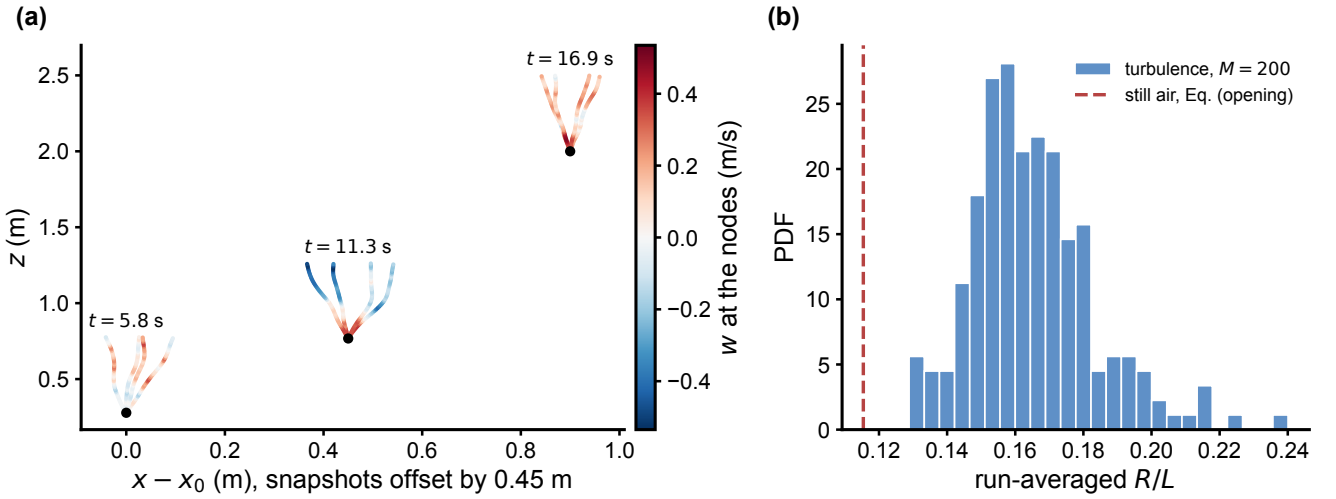



FIG. 8. Deformation of the bundle by turbulence ($N = 4$, $q = 0.331$ nC, $\bar{F}_l = 1$, $\sigma_w = 0.30$ m/s, $\ell = 1$ m). (a) One rising realization at three instants, side view in the frame of the spider (panels offset horizontally); nodes coloured by the vertical air velocity $w$ at the node. (b) Distribution over the $M = 200$ realizations of the time-averaged tip radius $R/L$, compared with the still-air value of Eq. (24) at $\bar{F}_l = 1$ ($\tau = 1$, $g = 1$).

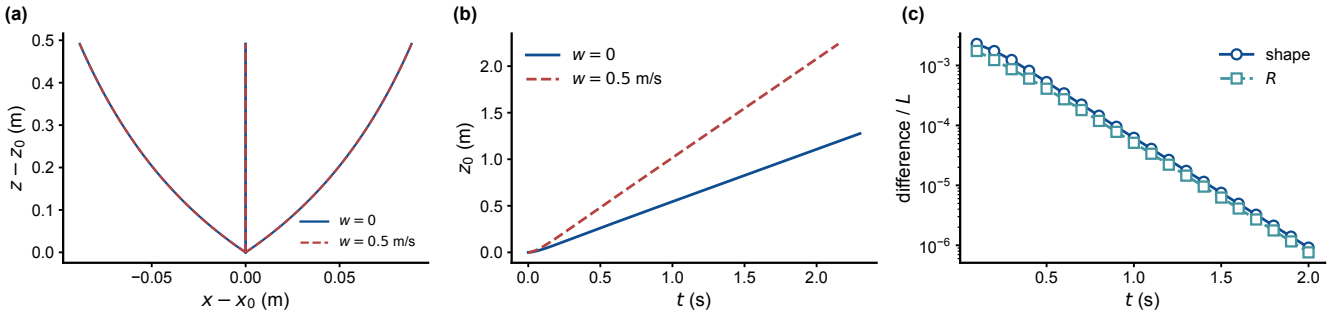



FIG. 9. Code test: shape invariance under a uniform updraft ($N = 4$, tip charge, $q = 0.662$ nC, $\bar{F}_l = 2$, $Q_s = 0$, constant $E = 7.41 \times 10^3$ V/m, $N_t = 100$). (a) Steady-state configurations in the frame of the spider for $w = 0$ (solid) and $w = 0.5$ m/s (dashed); the curves should coincide. (b) Ground-frame altitude $z_0(t)$; the slopes differ by exactly $w$. (c) Both runs started at rest in the ground frame: maximum shape difference and tip-radius difference, normalized by $L$; both decay exponentially with time constant 0.24 s.

*b. Total cost and break-even.* Let $c_{\text{sim}}$ and $c_{\text{sur}}$ be the costs of one estimate of $P$ with the simulator and with the surrogate, and $C_{\text{data}}$ and $C_{\text{train}}$ the one-time costs of generating the training data and training the network. The surrogate is cheaper overall after

$$Q^* = \frac{C_{\text{data}} + C_{\text{train}}}{c_{\text{sim}} - c_{\text{sur}}} \tag{39}$$

evaluations of $P$. Table XI gives the values, with costs counted in seconds of one vCPU (simulator, data) or of one GPU (surrogate, training); the two are not converted to a common monetary unit. With $M = 200$, one estimate of $P$ costs 38.7 vCPU-hours with the simulator (1.3 h of wall-clock time on 30 vCPUs) and 0.6–0.8 GPU-hours with the surrogate, and the one-time cost is recovered after about 21 estimates, i.e. about 4200 simulator runs.

*c. Accuracy at equal cost.* The break-even count is relevant only if the surrogate estimate of $P$ is accurate. The error of a surrogate estimate has a statistical part, $\propto M^{-1/2}$, and a systematic part (model bias) that does not decrease with $M$; the simulator has only the statistical part, with standard deviation $\sqrt{P(1-P)/M} \le 0.5/\sqrt{M}$. Already at $M = 10$ this bound is 0.16, below the mean $|\Delta P| = 0.45$–$0.46$ of the two GNS configurations on the interpolation set (Table X), and at $M = 10$ the simulator costs 1.9 vCPU-hours per point, about 4 min of wall-clock time on 30 vCPUs. At every budget that allows about ten simulator runs per point, the simulator is therefore the more accurate choice,

TABLE X. Surrogate accuracy on the test sets of Table V. One-step: mean squared error of the normalized acceleration on $10^5$ frames. Error at 1 s: root-mean-square position error of the thread nodes relative to the spider, over all runs, including unphysical ones. $|\Delta P|$: mean absolute error of the take-off probability over the grid points $(N, \sigma_w, x)$ of the test set, with the maximum in parentheses. The opening error $\Delta(R/L)$ is omitted, because the median time until a segment becomes unphysical is below 1 s for every method, shorter than the window over which the opening is measured. Speed-up: simulator wall time on a c2-standard-30 CPU over surrogate wall time, per test set; the learned models run on one NVIDIA L4 GPU and the two references on CPU, so the two are not directly comparable.

| Model | One-step MSE | Error at 1 s (mm) | $\lvert\Delta P\rvert$ | Physical at 1 s | Speed-up |
|---|---|---|---|---|---|
| *Interpolation ($N = 1$, 2, 4)* | | | | | |
| Constant velocity | 0.87 | 392 | 0.14 (0.45) | 0 | — |
| Follow-air | 1303 | 157 | 0.43 (0.90) | 0 | — |
| GNS-local | 0.087 | 34 | 0.46 (1.00) | 0.49 | 66 |
| GNS | 0.12 | 19 | 0.45 (1.00) | 0.27 | 47 |
| *Unseen $N$, interpolation ($N = 3$)* | | | | | |
| Constant velocity | 2.3 | 422 | 0.21 (0.32) | 0 | — |
| Follow-air | 1663 | 191 | 0.24 (0.46) | 0 | — |
| GNS-local | 0.25 | 45 | 0.36 (0.80) | 0 | 99 |
| GNS | 0.33 | 24 | 0.37 (0.80) | 0.03 | 67 |
| *Unseen $N$, extrapolation ($N = 6$, 8)* | | | | | |
| Constant velocity | 1.6 | 404 | 0.26 (0.63) | 0 | — |
| Follow-air | 1180 | 150 | 0.38 (0.70) | 0 | — |
| GNS-local | 18 | 371 | 0.52 (1.00) | 0 | 225 |
| GNS | 4.5 | 253 | 0.34 (0.56) | 0 | 152 |
| *Unseen $\sigma_w$ (0.22 m/s)* | | | | | |
| Constant velocity | 0.74 | 409 | 0.17 (0.28) | 0 | — |
| Follow-air | 929 | 144 | 0.25 (0.42) | 0 | — |
| GNS-local | 0.076 | 26 | 0.36 (0.82) | 0.30 | 93 |
| GNS | 0.12 | 16 | 0.38 (0.84) | 0.10 | 64 |

TABLE XI. Computational cost of the simulator (Algorithm 1) and of the surrogate on the interpolation test set ($N = 1$, 2, 4; $\sigma_w = 0.15$ and 0.30 m/s; 590 runs). Time per run: mean wall-clock time of one complete run. Time per $P$: $M = 200$ runs, in resource-hours (vCPU-hours for the simulator, GPU-hours for the surrogate). $C_{\text{data}}$: total simulator time of the training sweep; $C_{\text{train}}$: GPU time of training, including the validation rollouts. $Q^*$: Eq. (39).

| Method | Hardware | Time per run (s) | Time per $P$ (h) | $C_{\text{data}}$ (h) | $C_{\text{train}}$ (h) | $Q^*$ |
|---|---|---|---|---|---|---|
| Simulator | 1 vCPU, c2-standard-30 | 696 | 38.7 | — | — | — |
| GNS-local | 1 L4 GPU | 10.6 | 0.59 | 782 | 11.6 | 21 |
| GNS | 1 L4 GPU | 14.3 | 0.79 | 782 | 30.5 | 21 |

and the present surrogates do not pay back their one-time cost. The measured $|\Delta P|$ also contains the sampling error of the test estimates (10–20 runs per point on the interpolation set), which is below about 0.1 at every point and is the same for all methods; it does not change the comparison, since constant velocity reaches $|\Delta P| = 0.14$ on the same points.

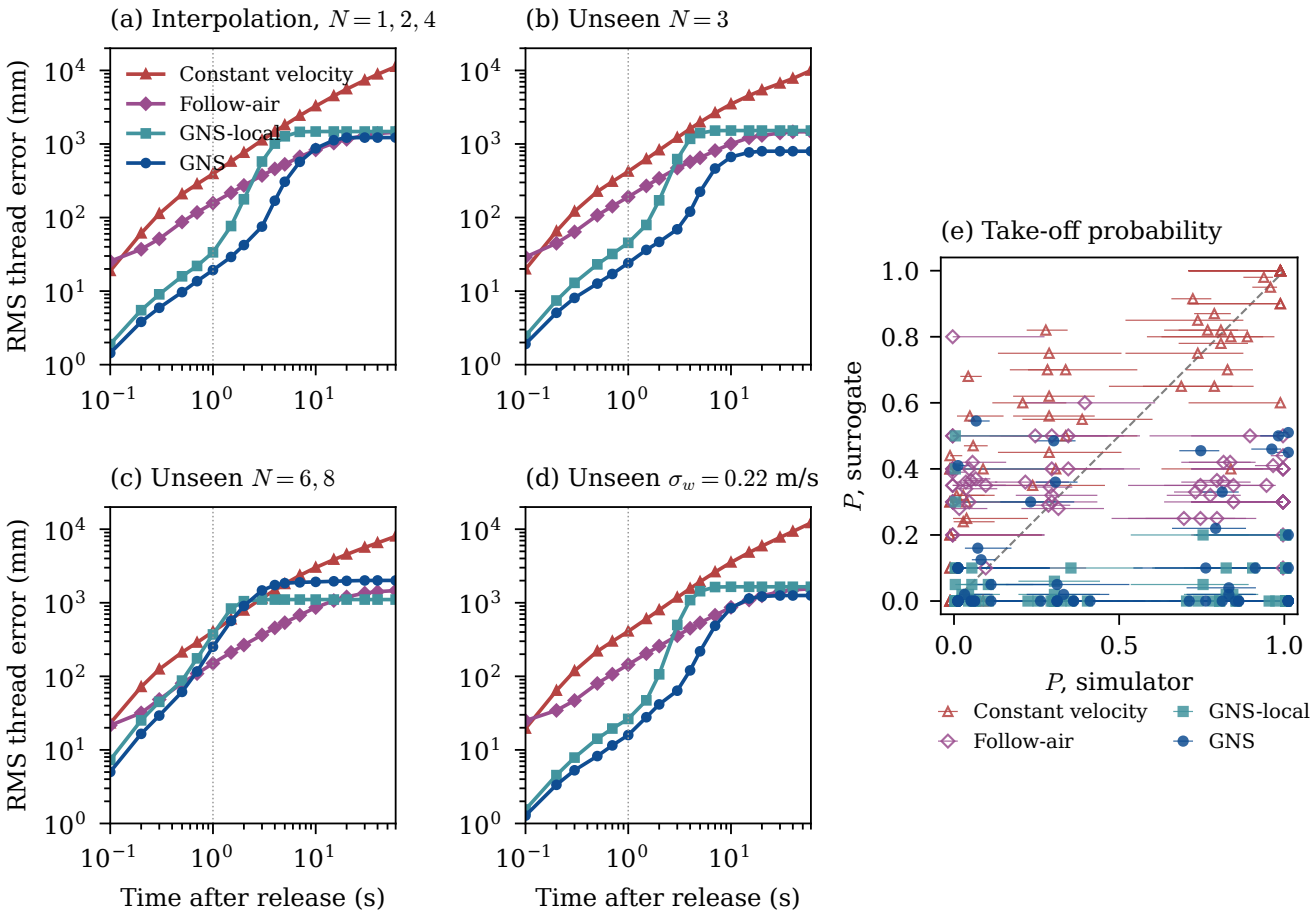



FIG. 10. Surrogate performance on the test sets of Table V. (a)–(d) Root-mean-square position error of the thread nodes relative to the spider versus time after release, averaged over all runs of the test set, for GNS, GNS-local, and the two reference methods; the dotted line marks 1 s. (a) Interpolation set ($N = 1$, 2, 4); (b) unseen $N = 3$; (c) unseen $N = 6$ and 8; (d) unseen $\sigma_w = 0.22$ m/s. (e) Take-off probability predicted by each method versus the simulator, at every grid point $(N, \sigma_w, x)$ of the four test sets; the dashed diagonal marks perfect agreement and the horizontal bars the Wilson 95% intervals of the simulator estimate. Points are offset horizontally by up to $\pm 0.012$ for readability.

## VIII. DISCUSSION

### A. What this work adds

Compared with [1], the present work (i) places the charged multi-thread bundle in realistic near-surface turbulence and treats take-off as a stochastic event; and (ii) gives an analytical drift–diffusion reference, Eq. (14), in which the electrostatic lift enters through $\bar{F}_l$ and the turbulence through a Péclet number. Turbulent ballooning has been simulated before only for single, uncharged filaments [23, 24]. Earlier work treated the two lift mechanisms mostly separately: aerodynamic models without charge [9, 10], and electrostatic models in still air [1, 2]. Field and laboratory studies indicate that both play a role [11, 12]. Equation (14) makes this quantitative: electrostatic lift shifts the drift $U_0$, while turbulence sets the width of the transition. The simulations (Sec. VII C) confirm the structure of Eq. (14) but not its parameters. For $N = 1$–8 and two turbulence intensities, $P$ depends only on the scaled drift $x$; the bundle enters only through $w_s$, so at fixed $\bar{F}_l$ adding threads lowers $w_s$, brings $x$ closer to zero, and widens the transition [Fig. 7(a)], while thread deformation and Coulomb opening have no measurable further effect. With $K = \sigma_w \ell$, Eq. (14) underestimates the steepness of the transition by a factor of about four and the probability at $x = 0$ by 0.09; both corrections are reproduced by massless tracers in the same fields and are therefore properties of the flow: the relevant diffusivity is the Lagrangian one, $\sigma_w^2 T_L$, and the finite correlation time raises the effective start fraction from $z_0/h = 0.25$ to 0.34. For predictions in the atmosphere, $T_L$ should be taken from surface-layer measurements rather

than from the synthetic field.

### B. What the learned surrogate adds

The GNS baseline answers a specific question: how far a generic learned simulator, which receives the air velocity and the charges as inputs but no force law, can replace the simulator for the quantity that matters here, the take-off probability. It improves the short-time trajectories: at 1 s after release the thread-node error of GNS is 19 mm on the interpolation set, against 157 mm for follow-air and 392 mm for constant velocity, so the network has learned the local response of the threads to drag, weight, and the electrostatic forces. It does not reproduce the take-off probability. The rollouts leave the physical manifold (a segment length off by more than 10%) within a median of 0.75 s, much shorter than a flight, and the accumulated error biases the outcome: $|\Delta P| = 0.45$ on the interpolation set, worse than the constant-velocity extrapolation (0.14), so the collapse of $P$ on $x$ in Fig. 7 is not recovered [Fig. 10(e)]. The model does not transfer to more threads than seen in training: on $N = 6$ and 8 the one-step error rises by a factor of about 40 and the rollouts become unphysical after 0.05 s. The speed-up of 47–225 per simulated second and the break-even after about 21 estimates of $P$ (Sec. VII D 1) would make the surrogate attractive for finer sweeps and inverse problems, but only once its bias is below the statistical error of a few simulator runs. The failure mode, drift of segment lengths, points to the missing structure: the network has to learn inextensibility and the Coulomb coupling from data, whereas a model that enforces the constraints of the rod by construction would remove the main source of error. This motivates the physics-structured surrogates of Sec. VIII F as the next step.

### C. Order-of-magnitude assessment of the mechanism

Over flat ground ($E_0 \approx 120$ V/m), a spider of 1 mg carrying a total charge of 1 nC receives an electrostatic force of $\sim 1.2\times10^{-7}$ N, about 1% of its weight ($\sim 10^{-5}$ N). Electrostatic lift alone therefore requires either a field enhanced by a prominence (grass tips, twigs), where $E$ reaches $10^4$–$10^5$ V/m [1, 3], or charges about two orders of magnitude larger than those measured. In the language of the model, take-off from flat ground is essentially aerodynamic ($\bar{F}_l \ll 1$, $w_c$ close to the settling speed), whereas take-off from an elevated point can be electrostatic or mixed. As the spider moves away from the prominence, $E$ decays towards $E_0$ and $\bar{F}_l$ drops. Electrostatic forces are therefore most relevant *at take-off and in the first centimetres to metres of flight*, and much less during long-distance transport. For the chamber profile of Eq. (1), the enhanced part of the field decays on the scales $z_1 = 1.5$ mm and $z_2 = 7.9$ mm: at $z = 3$ cm above the prominence it is below 2% of $E_\infty$, so a bundle with $\bar{F}_l$ close to 1 at the prominence loses most of its electrostatic lift within the first centimetres of flight, a distance much shorter than the thread length.

### D. Potential applications

- **Dispersal ecology.** Combined with routinely measured meteorological quantities ($u_*$, and hence $\sigma_w$; atmospheric potential gradient), the take-off probability predicts under which weather conditions ballooning is likely, and how it depends on spider mass and number of threads.
- **Benchmark for learned simulators.** The simulations form a controlled data set with known analytical limits, suitable for testing graph-based surrogate models of filament dynamics with long-range interactions [15, 20].

### E. Limitations

*a. Of the model.*

- RFT neglects hydrodynamic interactions between threads, although the threads are very close near the spider [1]. A slender-body treatment would reduce the drag of the bundle near the spider.
- The charge distribution along the thread is inferred from the motion of the spider, not measured [3]; only its tip-charge idealization is treated.
- All threads start from the centre of the spider, which is a point mass without rotation, and are separate from the base on; in reality they leave the spinnerets at the body surface ($r_s \approx 1$ mm, 0.2% of $L$) and may be joined near the spinnerets.

- The charge is constant in time. Silk conductivity increases roughly exponentially with relative humidity [13], so in humid air the charge may leak within the flight time.
- The turbulence is synthetic, homogeneous, and isotropic, with statistics matched only for the vertical component; the growth of the eddy size with height, mean shear, and convective plumes are not included.
- Values of $w_c$ above $\sim 1$ m/s (Table II) lie outside the range of validity of the low-Reynolds-number drag laws.
- The drift–diffusion reference (14) assumes a height-independent diffusivity; near the ground $K$ grows with $z$.
- The electric field is uniform in height in the turbulence ensembles, so that $\bar{F}_l$ and the drift $U_0$ are constant during the flight. Near a prominence the enhanced field decays with height over a distance comparable to the size of the prominence, and $\bar{F}_l$ would decrease as the spider rises.
- The fitted Lagrangian correction ($a \approx 4$, $p_0 \approx 0.34$) is a property of the kinematic-simulation field, which does not sweep small eddies with the large ones; the collapse on $x$ is expected to hold more generally, but the values of $a$ and $p_0$ need not carry over to real surface-layer turbulence.

*b. Of the mechanism.* The electrostatic mechanism depends on field enhancement near objects and on nanocoulomb charges whose origin (triboelectric charging during spinning, or induction) is not established. The measured charges [3] are indirect, inferred from the motion in a known field. The mechanism thus explains take-off in calm air from elevated points, but it cannot on its own explain long-distance transport at altitudes of hundreds of metres.

### F. Future work

1. Anisotropic, height-dependent surface-layer turbulence (e.g. from large-eddy simulation), mean shear, and convective plumes.
2. Time-dependent charge, with a leakage rate depending on humidity [13].
3. Hydrodynamic interactions between threads (slender-body theory or regularized Stokeslets).
4. Physics-structured surrogates, in which the known force laws (resistive drag, Coulomb repulsion, gravity, electrostatic lift) are evaluated exactly and the network learns only the internal response of the threads, trained for stability over long rollouts and checked against the analytical results of Secs. III–IV.

## IX. CONCLUSIONS

We placed the charged multi-thread ballooning model of Habchi and Jawed [1] in synthetic near-surface turbulence with surface-layer statistics. A uniform flow leaves the bundle unchanged, so the relevant effect of the air is its spatial and temporal variability on the scale of the bundle. Take-off then becomes a stochastic event; in the limit of large, slow eddies its probability follows a drift–diffusion law that smooths the deterministic threshold $\bar{F}_l = 1$ into a sigmoid set by a Péclet number. The simulations reproduce the validation cases of [1] within 4.4% (the target read from a figure) and analytical terminal velocities within 0.08%; for tip-charged threads in still air they satisfy the exact first integral $T^\beta \sin\theta = \text{const.}$ and the small-angle opening law within 2% for $R/L \leq 0.2$. In turbulence, the take-off probability of 8200 simulated bundles with $N = 1$–$8$ collapses on a single function of the scaled drift $x = w_s(\bar{F}_l - 1)h/K$: the bundle enters only through its settling speed. The drift–diffusion law has the right form, but its width is set by the Lagrangian diffusivity of the flow, about four times smaller than $\sigma_w \ell$ in the synthetic field, as confirmed with massless tracers. An order-of-magnitude assessment indicates that electrostatic lift matters at take-off from elevated points, whereas over flat ground the aerodynamic mechanism dominates. Finally, a graph network simulator trained on bundles with $N = 1$, 2, and 4 threads predicts the take-off probability for $N = 3$ and for $N = 6$ and 8 with mean absolute errors of 0.37 and 0.34, at a cost 67 and 152 times lower per simulated second than the full simulation; a constant-velocity extrapolation gives 0.21 and 0.26, so the generic network does not yet replace the simulator for $P$.

### Appendix A: Resistive force coefficients

For a slender filament of length $L$ and radius $r \ll L$ at low Reynolds number, the drag coefficients per unit length of Eq. (3) follow from slender-body theory to leading order in $1/\ln(L/r)$.

The coefficients of Eq. (3) are the leading-order slender-body (Cox) coefficients for a straight cylinder of length $L$ and radius $r$; they appear in exactly this form in [6] and derive from [5]. Reynolds number of the thread, $\mathrm{Re} = \rho_{\mathrm{air}} v r/\mu$ ($\rho_{\mathrm{air}} = 1.2$ kg/m$^3$): 0.006 at $v = 0.29$ m/s ($N = 8$), 0.04 at 2.2 m/s ($N = 1$), 0.4 at 21.7 m/s ($m = 10$ mg, $N = 1$); RFT is valid except at the largest $w_c$ of Table II. The spider body is the weaker point: its Reynolds number $\rho_{\mathrm{air}} v\, 2r_s/\mu$ is 38 at 0.29 m/s and 280 at 2.2 m/s, so Stokes drag underestimates its drag by a factor of about 3 and 8 (Schiller–Naumann correction $1 + 0.15\,\mathrm{Re}^{0.687}$). Because $\zeta_s \ll N\eta_\parallel L$, this changes $w_s$ by about 2% for $N = 8$ but by about 40% for $N = 1$ at $m = 1$ mg.

### Appendix B: Finite-angle correction to the opening law

Equation (24) uses $\sin\theta \approx \tan\theta \approx \theta$.

In the limit $\tau \to 1$ ($U \to 0$) the threads are straight and the tip balance is exact: $R/L = \sin\theta_L$ and $\tan\theta_L = F_C/(qE) = S_N\Lambda/[4(R/L)^2]$. With $\rho = R/L$ and $\rho_0^3 = S_N\Lambda/4$ (Eq. (24) at $g = 1$), $\tan\theta_L = \rho(1 + \rho^2/2 + \dots)$ gives $\rho^3(1 + \rho^2/2) = \rho_0^3$, i.e.

$$\frac{R}{L} = \rho_0 \left(1 - \frac{\rho_0^2}{6}\right) + O(\rho_0^5).$$

The small-angle law overestimates $R$ by $\rho_0^2/6$: 0.3% for $N = 2$ and 1.5% for $N = 8$ at $\Lambda = 10^{-2}$. Check against the clamped still-air runs ($\bar{F}_l = 1$): predicted 0.14, 0.22, 0.29%, simulated 0.12, 0.20, 3.2% for $N = 2$, 4, 8. The correction explains $N = 2$ and 4; the 3% for $N = 8$ is not a small-angle effect (possibly tips not exactly on one circle, or interactions near the spider) and remains open. For $\tau < 1$ the curvature of the thread adds corrections of the same order through $g(\tau)$.

### Appendix C: Simulation parameters

Table XII lists the numerical parameters of Algorithm 1 used in Secs. VII A–VII B; the turbulence ensembles differ only in $N_t = 50$ and the hardware (Appendix E).

TABLE XII. Numerical parameters used in all simulations.

| Parameter | Symbol | Value |
|---|---|---|
| Edges per thread | $N_t$ | 100 (convergence study: 25–200) |
| Initial / maximum / minimum time step | $\Delta t_0$, $\Delta t_{\max}$, $\Delta t_{\min}$ | $10^{-4}$ s, $10^{-2}$ s, $10^{-9}$ s |
| Newton tolerance | $\epsilon$ | $10^{-10}$ N (on $\|\boldsymbol{F}\|_1$) |
| Maximum Newton iterations | $K$ | 20 |
| Steady-state tolerance and window | $\delta$, $t_w$ | $10^{-6}$, 0.05 s |
| Clamp time before release (Alg. 2) | $t_c$ | 0.05 s (sweep), 3 s (check) |
| Initial thread spacing | $d_0$ | 100 $\mu$m |
| Poisson ratio; twist inertia | $\nu$; — | 0.5; $\rho_t J\, l_0/2$ |
| Tangent dependence of $\mathbf{C}$ in $\mathbf{J}$ | — | lagged |
| Output interval | $\Delta t_{\text{out}}$ | $10^{-2}$ s ($10^{-3}$ s for test 1) |
| Coulomb softening length | — | 0 (off) |
| Spider charge as Coulomb source for thread nodes | — | no, as in [1] |
| Code and version | — | Python 3.14, NumPy 2.5, SciPy 1.18 |
| Hardware | — | Apple M4 Pro, 10 performance cores, CPU only |

### Appendix D: Grid of the turbulence ensembles

Table XIII gives, for every grid point of the sweep of Sec. VII C, the lift-to-weight ratio and the charge per thread that follow from the prescribed scaled drift $x$.

TABLE XIII. Lift-to-weight ratio $\bar{F}_l$ and charge per thread $q$ (nC, in parentheses) at each grid point of Algorithm 2, from $\bar{F}_l = 1 + x\sigma_w \ell/(w_s h)$ and $q = \bar{F}_l\, mg/(NE)$, with $m = 1$ mg, $E = 7.41 \times 10^3$ V/m, $\ell = 1$ m, $h = 2$ m. The settling speed $w_s$ is that of Eq. (10). Dashes: points with $\bar{F}_l \le 0$, excluded from the analysis (Sec. VII C); blank: not simulated.

| $N$ | $\sigma_w$ (m/s) | $w_s$ (m/s) | $x = -4$ | $-2$ | $-1$ | 0 | 1 | 2 | 4 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.15 | 2.170 | 0.862 (1.141) | 0.931 (1.232) | 0.965 (1.278) | 1.000 (1.324) | 1.035 (1.370) | 1.069 (1.415) | 1.138 (1.507) |
| 1 | 0.30 | 2.170 | 0.724 (0.958) | 0.862 (1.141) | 0.931 (1.232) | 1.000 (1.324) | 1.069 (1.415) | 1.138 (1.507) | 1.276 (1.690) |
| 2 | 0.15 | 1.128 | 0.734 (0.486) | 0.867 (0.574) | 0.934 (0.618) | 1.000 (0.662) | 1.066 (0.706) | 1.133 (0.750) | 1.266 (0.838) |
| 2 | 0.22 | 1.128 | | | 0.903 (0.597) | 1.000 (0.662) | 1.097 (0.726) | | |
| 2 | 0.30 | 1.128 | 0.468 (0.310) | 0.734 (0.486) | 0.867 (0.574) | 1.000 (0.662) | 1.133 (0.750) | 1.266 (0.838) | 1.532 (1.014) |
| 3 | 0.30 | 0.762 | | | 0.803 (0.354) | 1.000 (0.441) | 1.197 (0.528) | | |
| 4 | 0.15 | 0.576 | 0.479 (0.158) | 0.739 (0.245) | 0.870 (0.288) | 1.000 (0.331) | 1.130 (0.374) | 1.261 (0.417) | 1.521 (0.503) |
| 4 | 0.22 | 0.576 | | | 0.809 (0.268) | 1.000 (0.331) | 1.191 (0.394) | | |
| 4 | 0.30 | 0.576 | — | 0.479 (0.158) | 0.739 (0.245) | 1.000 (0.331) | 1.261 (0.417) | 1.521 (0.503) | 2.042 (0.676) |
| 6 | 0.30 | 0.386 | | | 0.612 (0.135) | 1.000 (0.221) | 1.388 (0.306) | | |
| 8 | 0.15 | 0.291 | — | 0.484 (0.080) | 0.742 (0.123) | 1.000 (0.165) | 1.258 (0.208) | 1.516 (0.251) | 2.032 (0.336) |
| 8 | 0.30 | 0.291 | — | — | 0.484 (0.080) | 1.000 (0.165) | 1.516 (0.251) | 2.032 (0.336) | 3.064 (0.507) |

### Appendix E: Reproducibility of the ensemble simulations

The turbulence ensembles (Sec. VII C) use $N_t = 50$ edges per thread; for test 2 this changes $\bar{v}_t$ by 0.5% and $R/L$ by 0.13% relative to $N_t = 100$ (Table VIII). They were run with one fixed version of the code on two Google Cloud

`c2-standard-30` instances, with resume from per-run outputs. Before production we checked: (i) the synthetic field, renormalized so that the sample standard deviation of $w$ equals $\sigma_w$ (correction factor 1.019), with the spectrum of Fig. 6; (ii) that the trajectories do not depend on the mesh through the turbulence ($k_{\max}$ fixed at $2\pi/(5 \times 5\text{ mm}) = 251\text{ m}^{-1}$; spider positions for $N_t = 50$ and 100 within 0.6 cm after the same flight); (iii) the time-step cap $\Delta t_{\max} = 10^{-2}$ s (difference 0.65 cm against $\Delta t_{\max} = 10^{-3}$ s); (iv) that a run interrupted and resumed reproduces the uninterrupted one; and (v) that massless tracers in the same fields give the same take-off probability at $x = 0$ as the bundles. A regression test compared the production code with the version used for Secs. VII A–VII B: on the same machine the outputs are identical, and between the cloud and the computer of Table XII the largest relative difference is $1.4 \times 10^{-6}$ (in $\theta_L$), apart from quantities at round-off level. Newton's method failed to converge at 5 time steps in total over the 8200 valid runs, all of which completed. Each run is stored as an HDF5 file (Table IV) with the code version, the seed, and all parameters as attributes. For the learned surrogate, the HDF5 files are converted once into a preprocessed data set, which contains the splits of Table V, the training statistics, and the air velocity at the current positions and the next time; the conversion was checked field by field against the HDF5 files (maximum difference zero). The air velocity in the rollouts is computed from the stored modes, which are in single precision; for $|\boldsymbol{u}| > 0.01$ m/s it agrees with the simulator to a relative difference of $4.5 \times 10^{-4}$. A provenance record kept with the code links every number reported in Secs. VII C–VII D 1 to the code version and the runs that produced it.

## Appendix F: List of symbols

TABLE XIV. Main symbols.

| | | | |
|---|---|---|---|
| $N$ | number of threads | $\eta_\parallel, \eta_\perp$ | RFT drag coefficients |
| $L$, $r$ | thread length, radius | $\beta$ | $\eta_\perp/\eta_\parallel$ |
| $m$, $r_s$ | spider mass, radius | $\zeta_s$ | Stokes drag coefficient of the spider |
| $q$ | charge per thread | $T(s)$ | tension |
| $E(z)$ | atmospheric electric field | $\theta(s)$ | angle from the vertical |
| $\boldsymbol{u}$, $w$ | air velocity, updraft speed | $R$ | radius of the tip circle |
| $U$ | velocity relative to air | $S_N$ | Coulomb sum, Eq. (19) |
| $V$ | ground-frame velocity | $\tau$ | tension ratio $T_0/T_L$ |
| $\bar{F}_l$ | lift-to-weight ratio | $\Lambda$ | Coulomb-to-lift ratio |
| $x$ | scaled drift | $w_c$, $w_s$ | mean-flow threshold, settling speed |
| $\sigma_w$, $\ell$ | turbulence intensity, scale | $P$, $K$ | take-off probability, diffusivity |


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